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(54) **DISPLAYING APPLICATION VIEWS**

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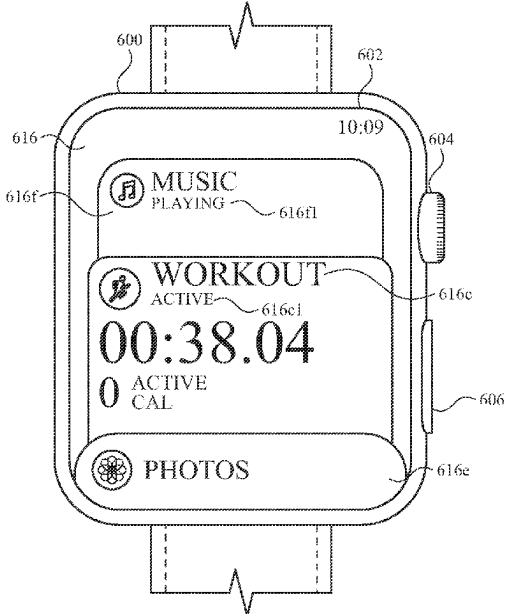
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(57) **ABSTRACT**

A method includes receiving, a request to display a set of application views. In response to receiving the request to display a set of application views, a set of application views is displayed. The set of application views includes a first application view that corresponds to a first application that is not in an active session, and the set of application views are arranged in an order of application views. Displaying the set of application views includes: displaying a second application view that corresponds to the second application before the first application view in the order of application views in accordance with a determination that a second application is in an active session; and displaying the first application view in a first position in the order of application views in accordance with a determination that there are no applications in an active session.

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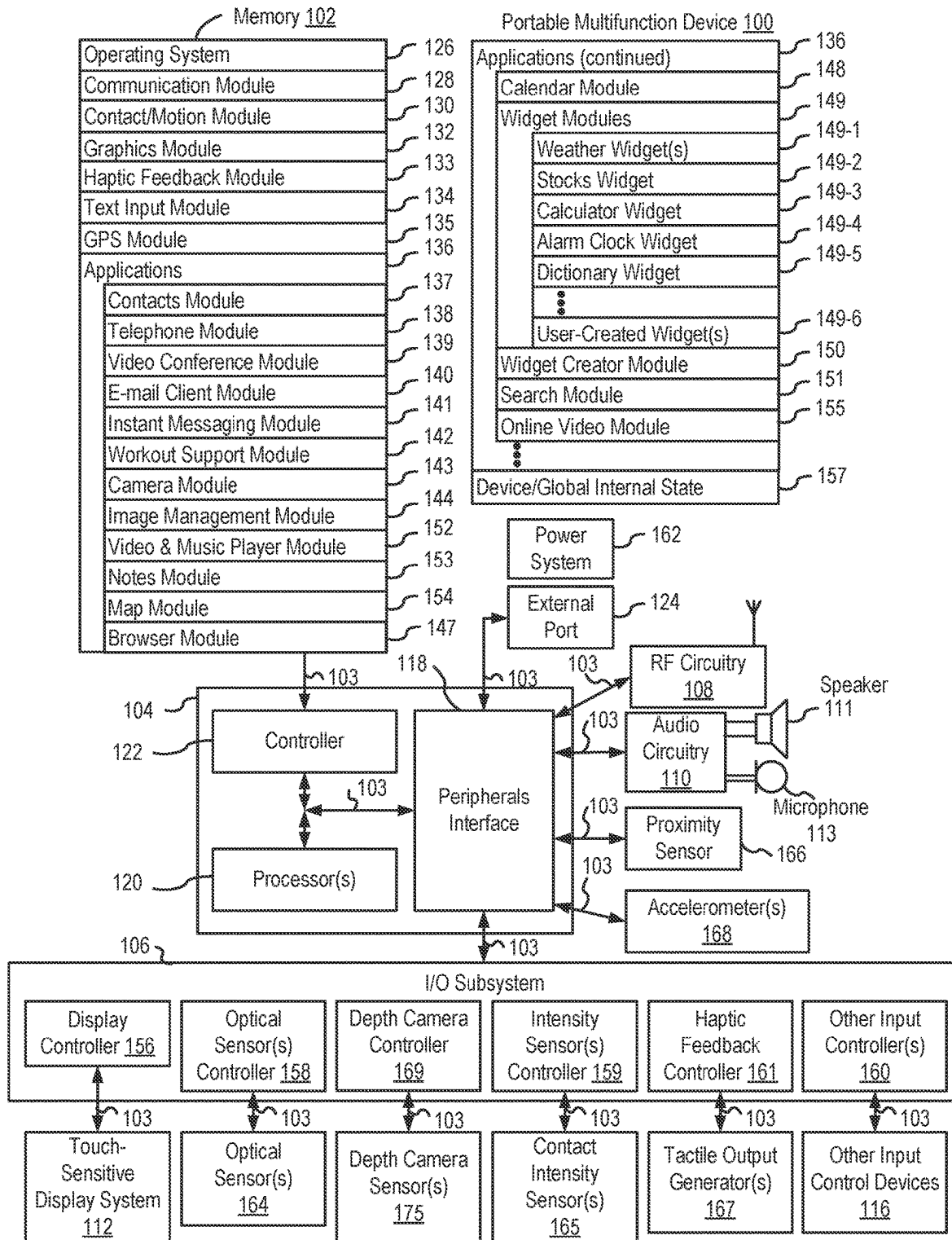


FIG. 1A

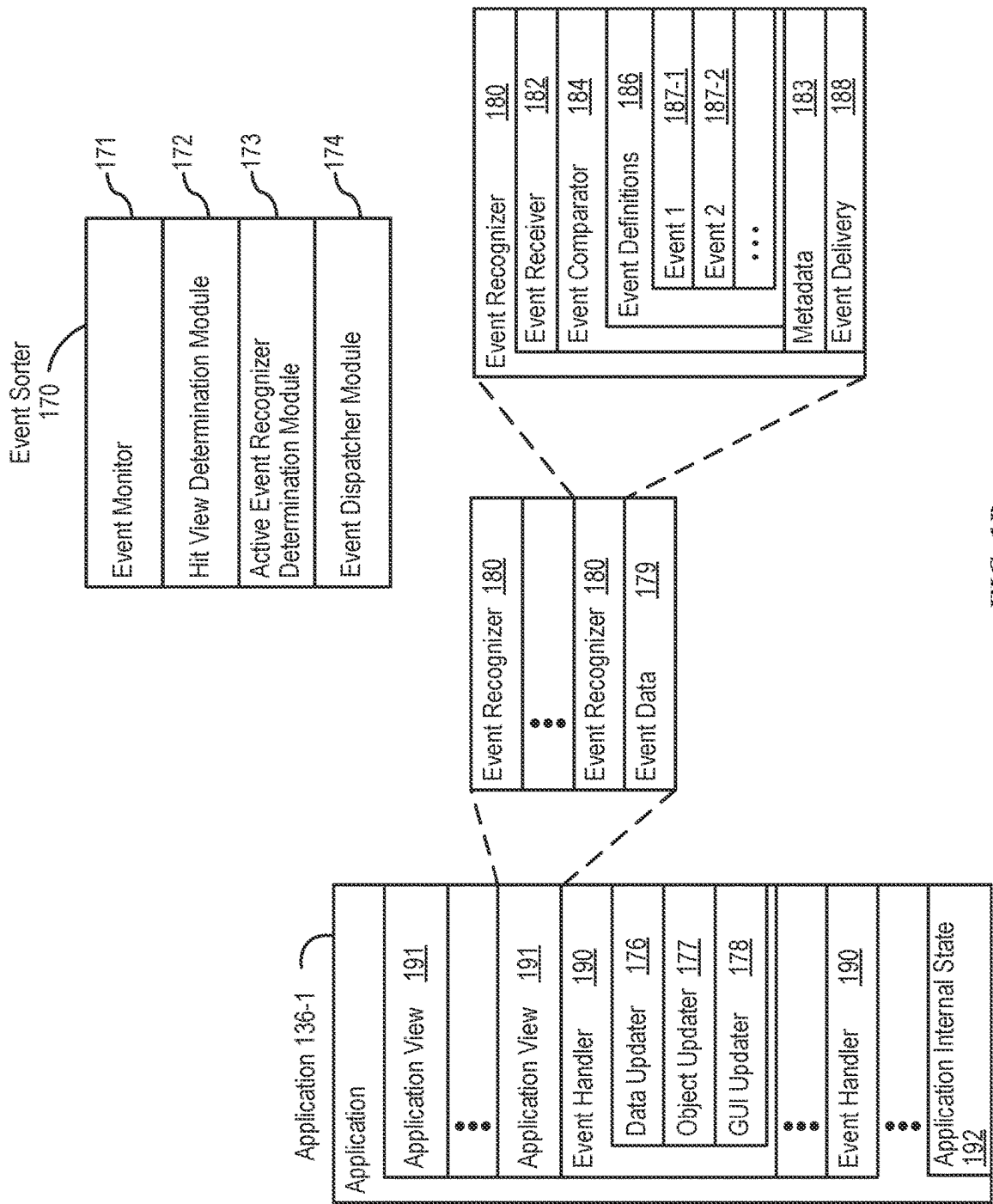


FIG. 1B

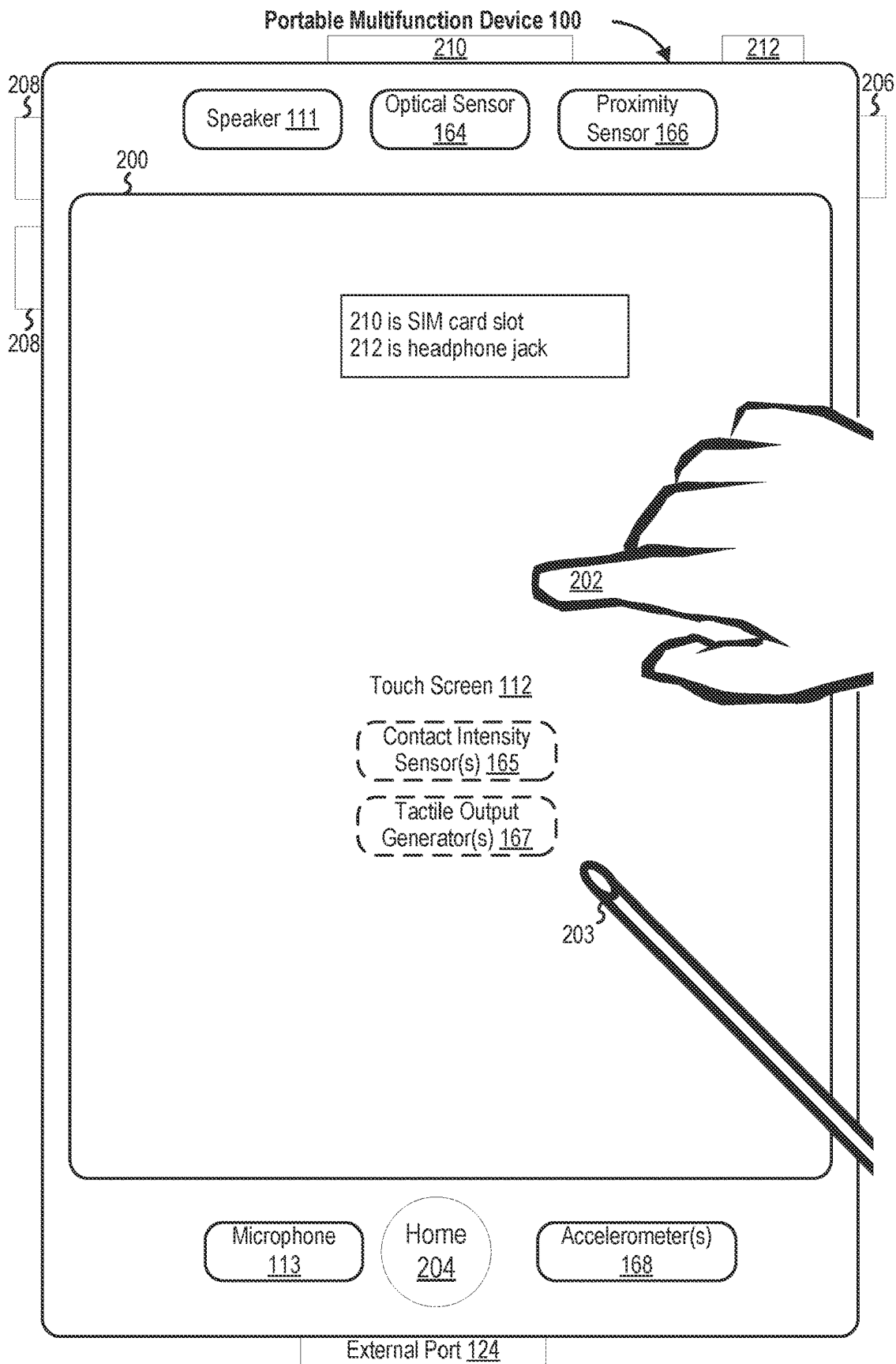


FIG. 2

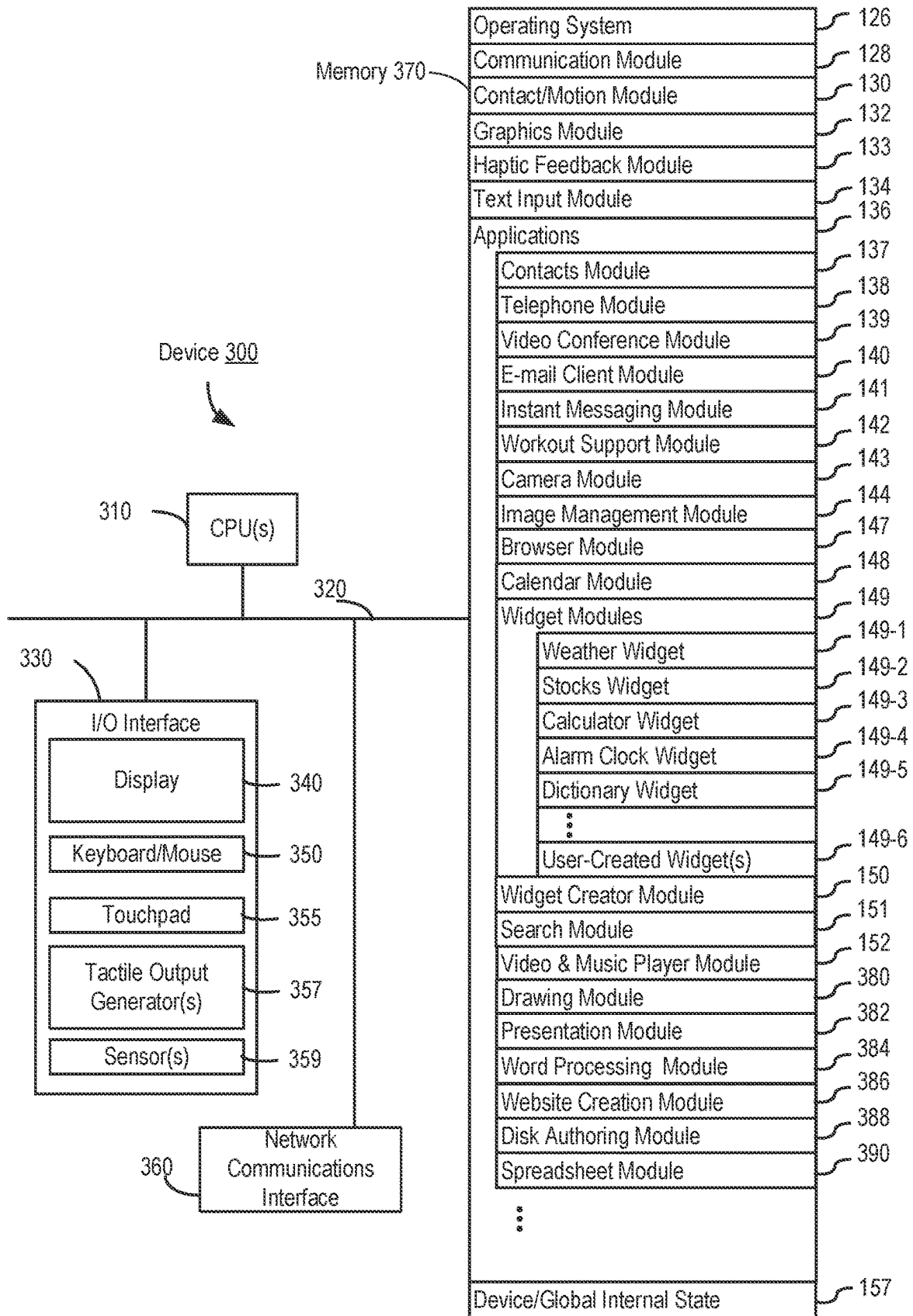


FIG. 3

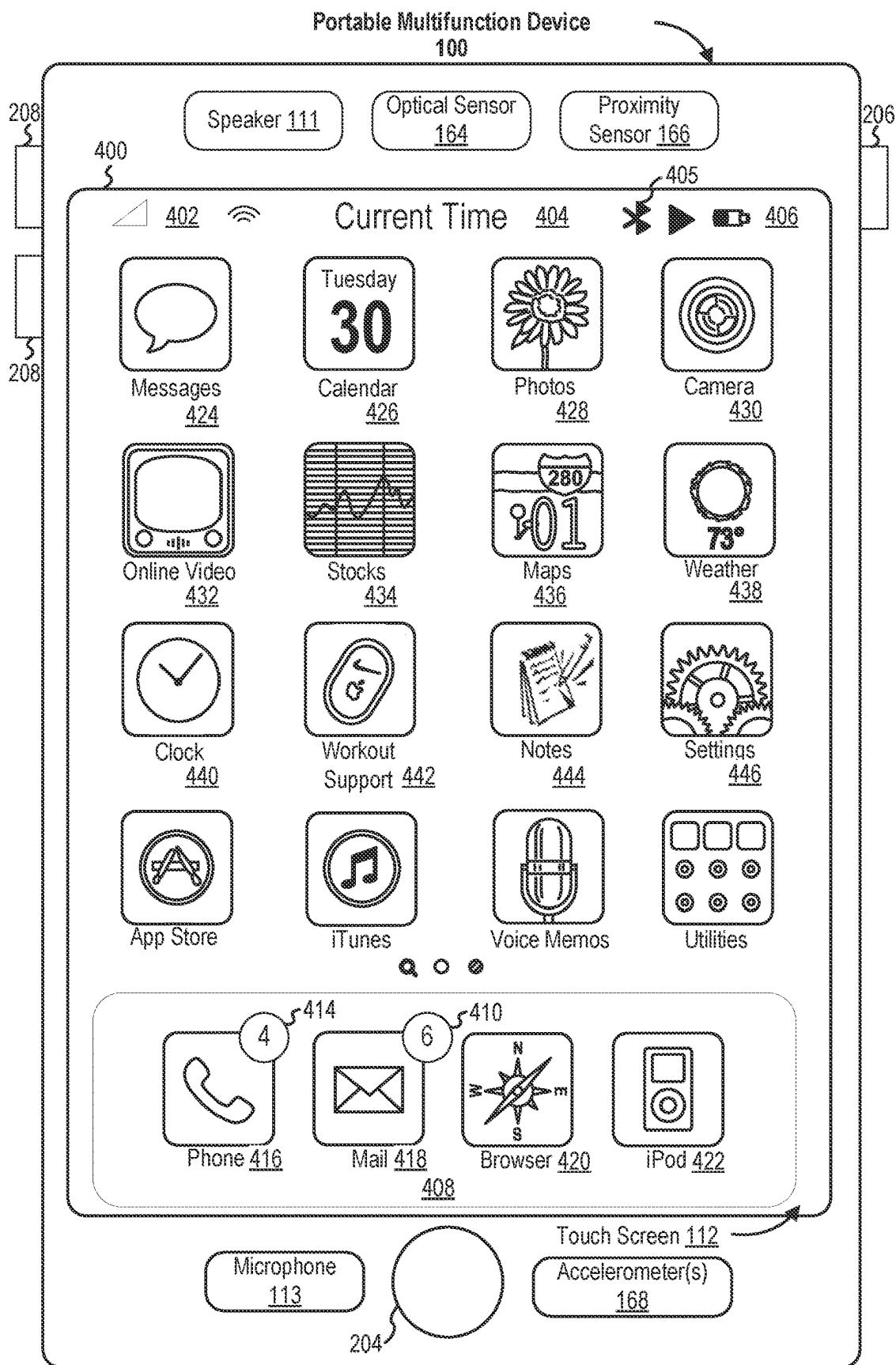


FIG. 4A

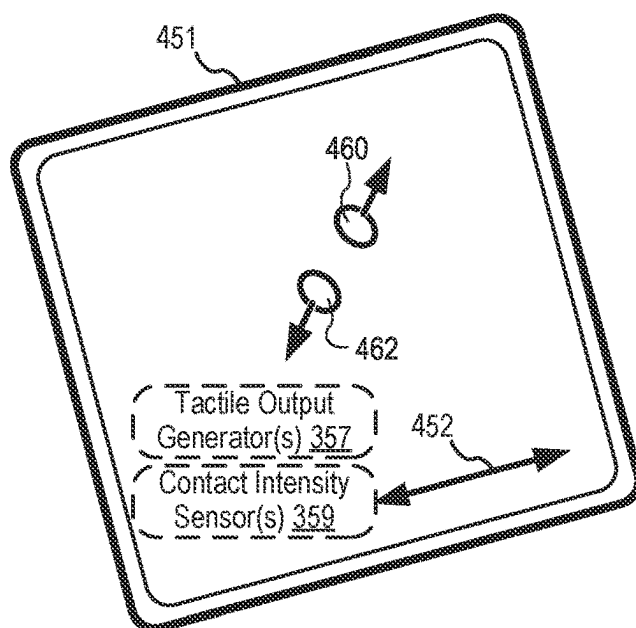
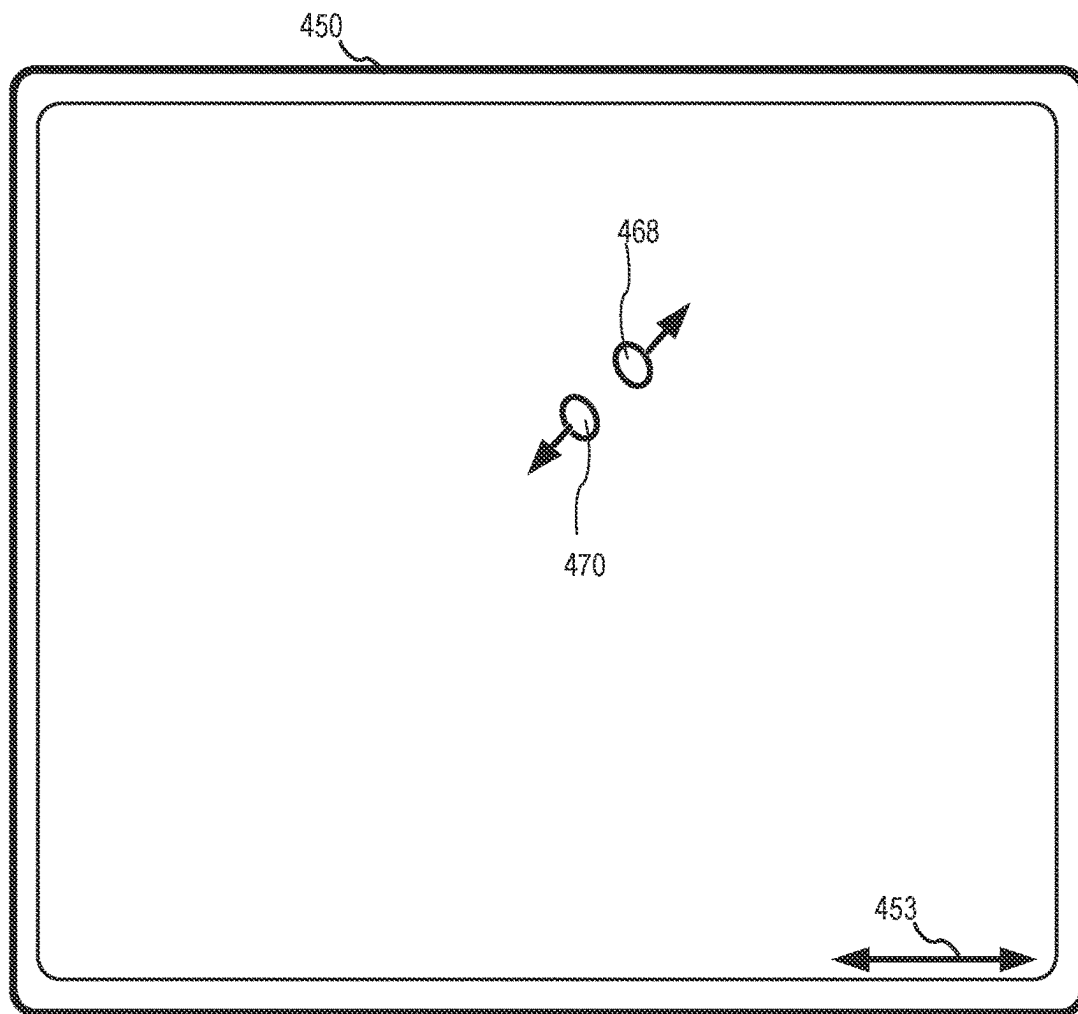
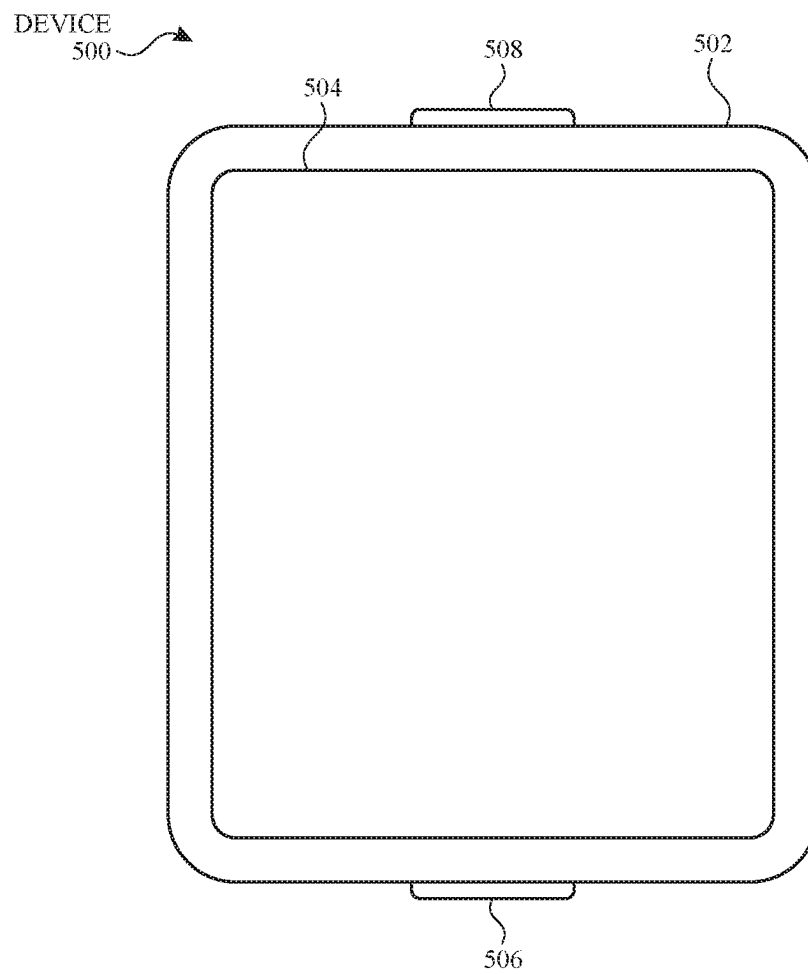


FIG. 4B

*FIG. 5A*

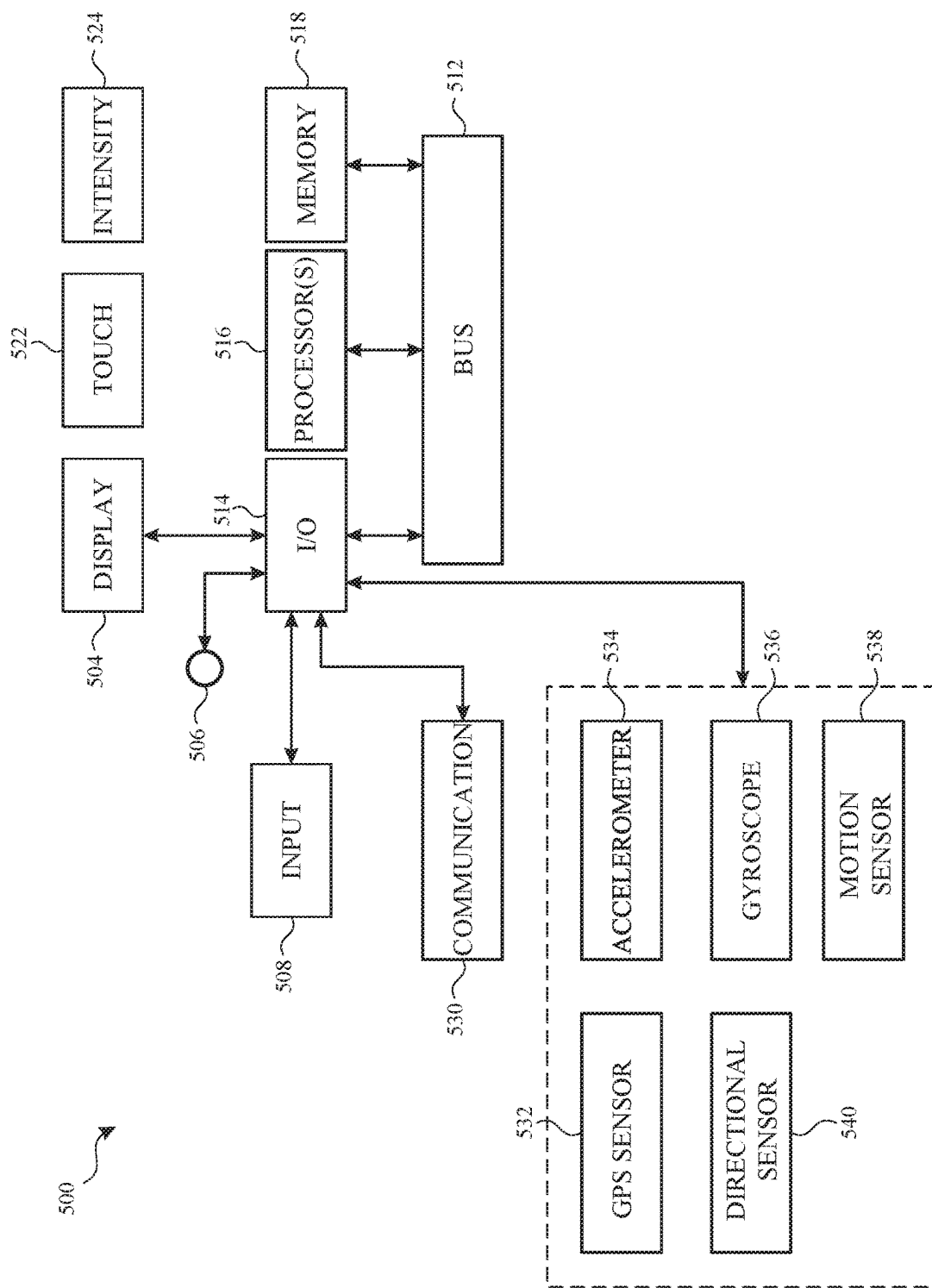


FIG. 5B

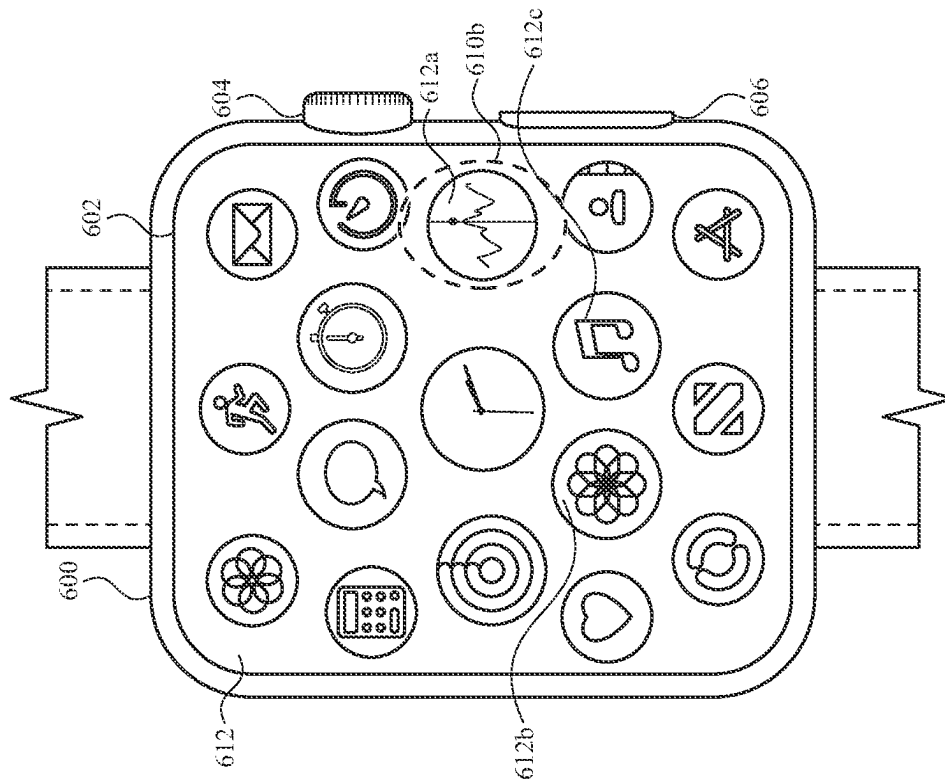


FIG. 6B

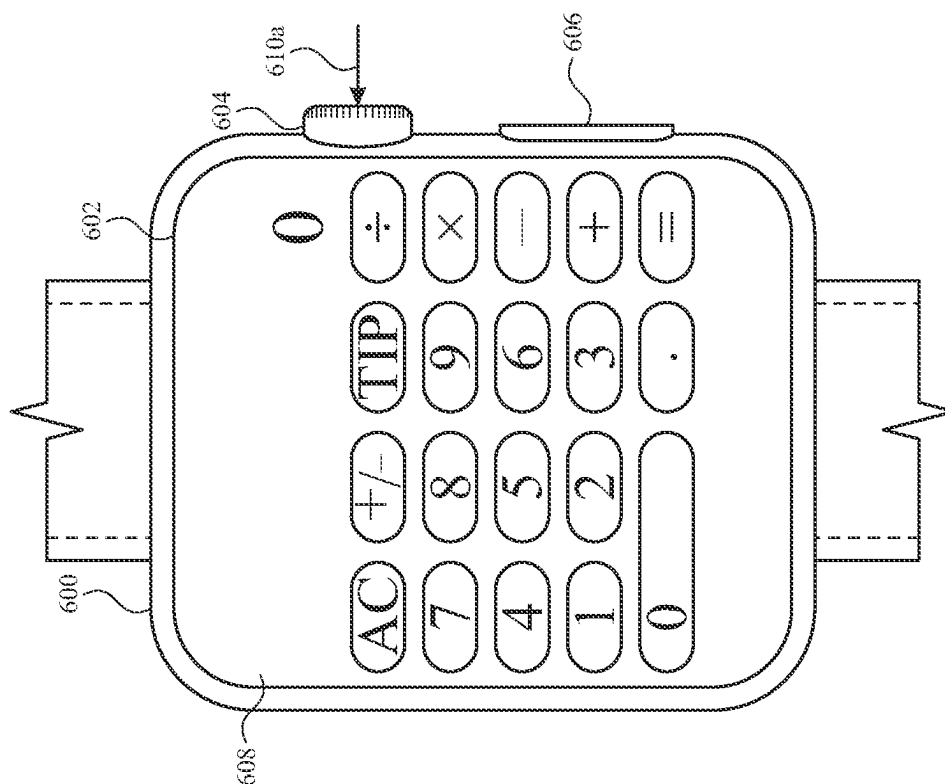


FIG. 6A

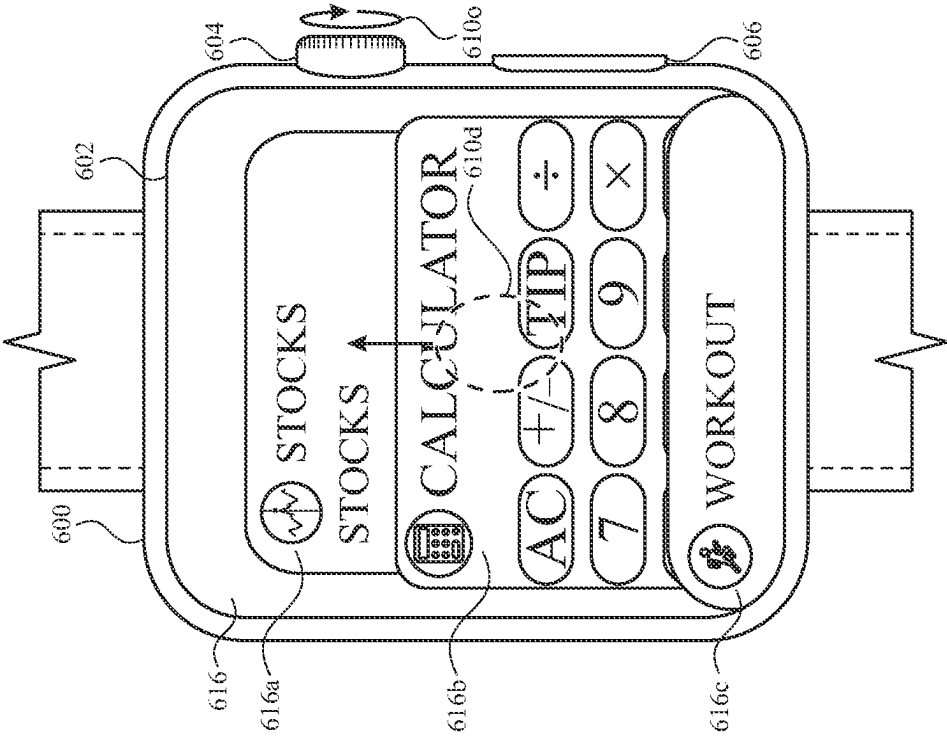


FIG. 6D

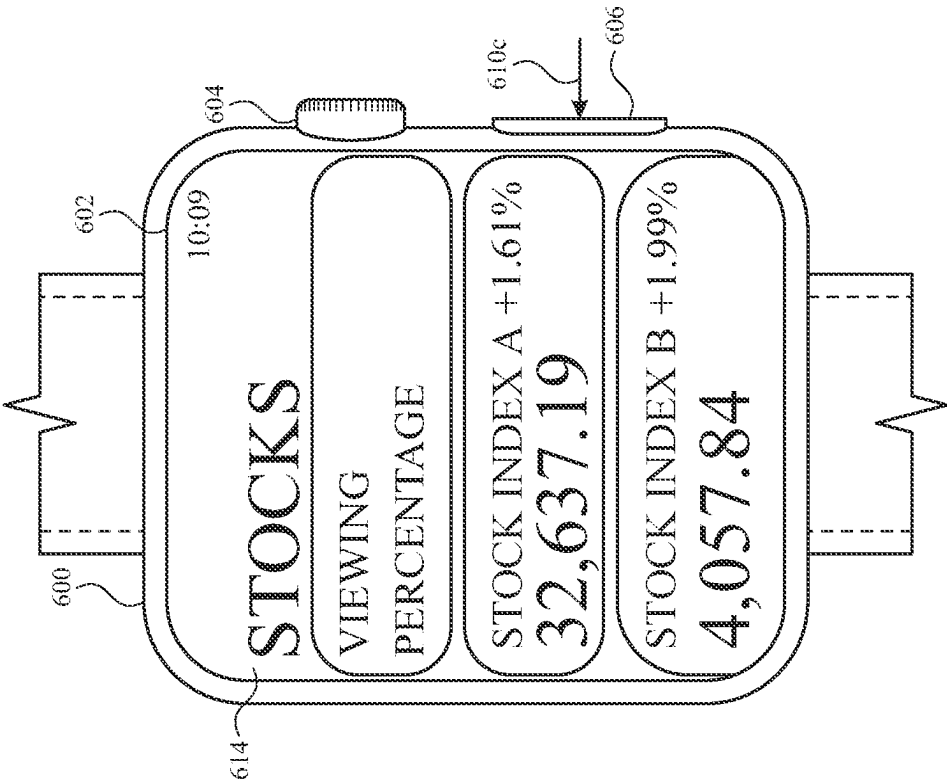


FIG. 6C

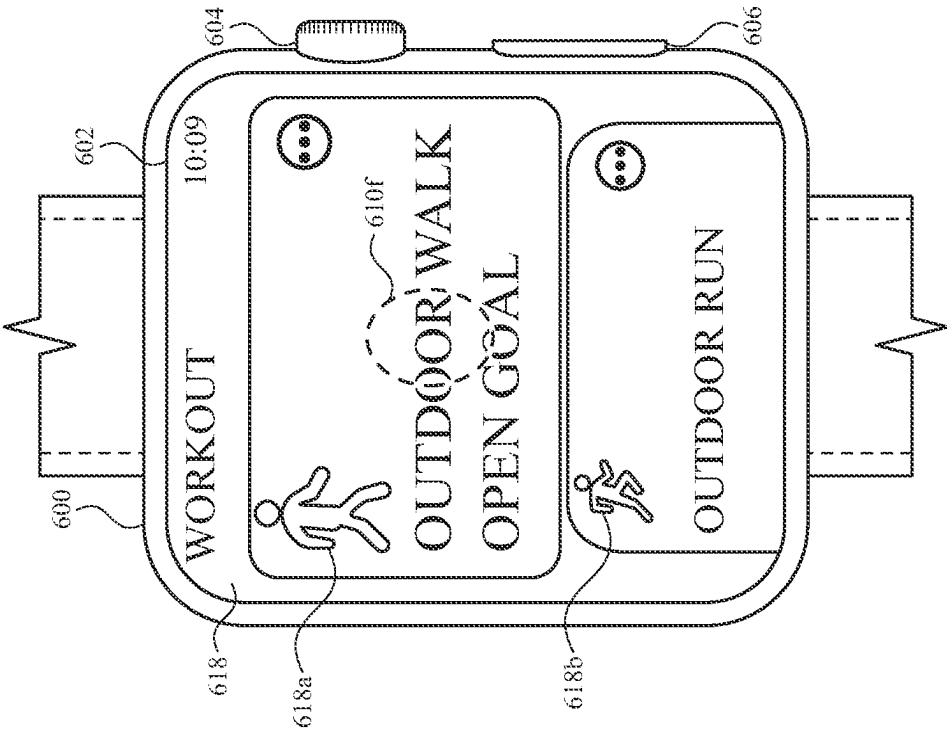


FIG. 6F

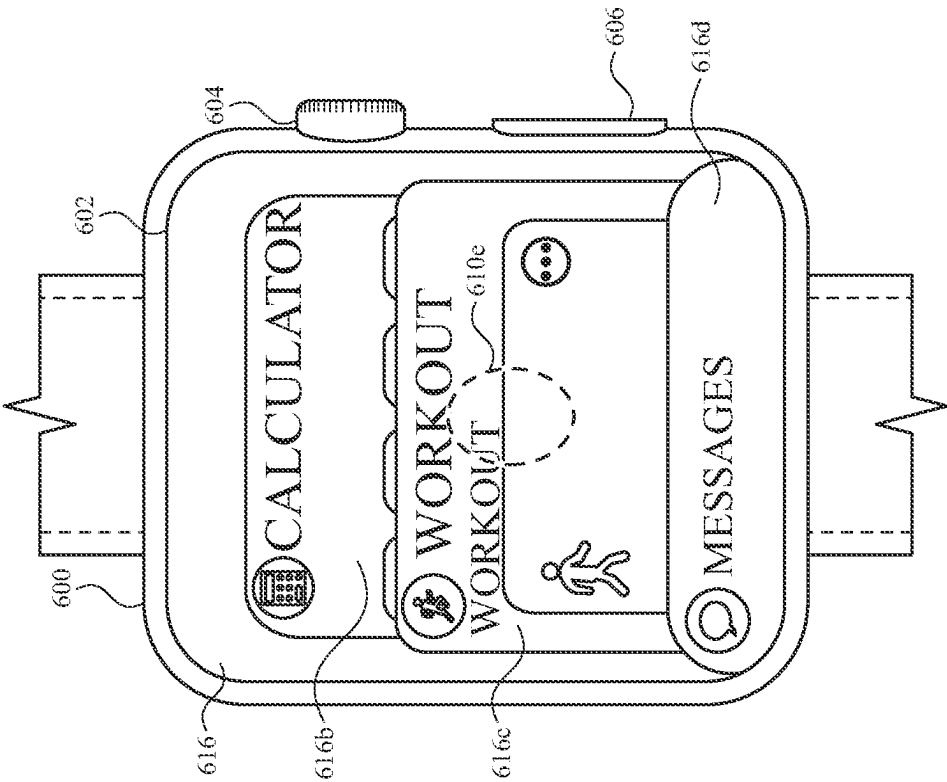
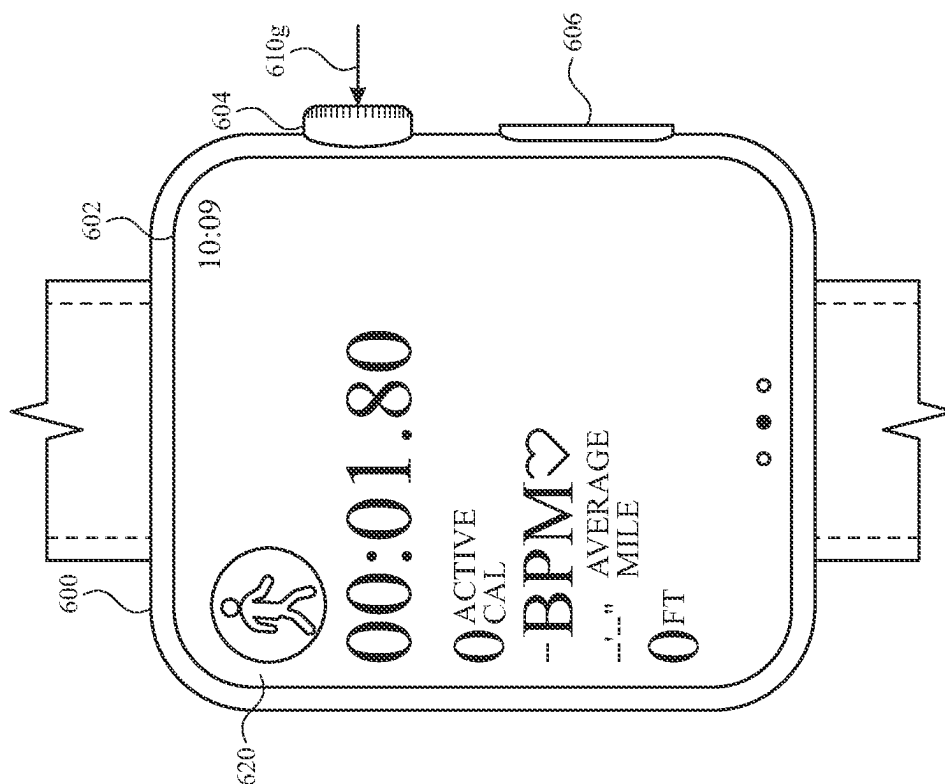
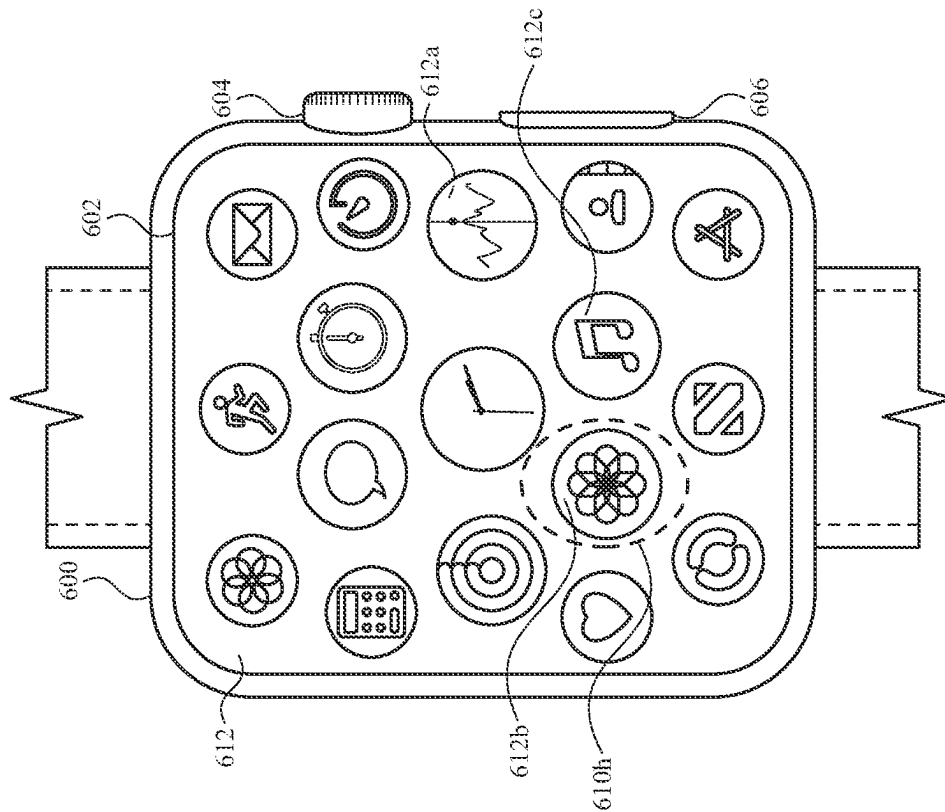


FIG. 6E



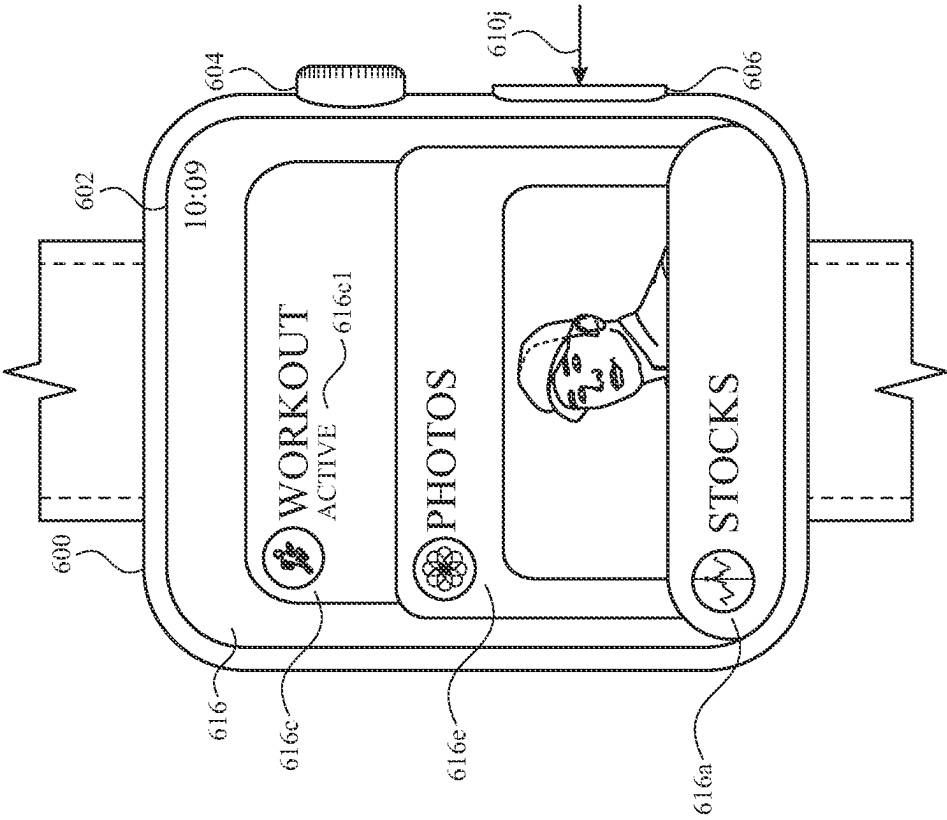


FIG. 6J

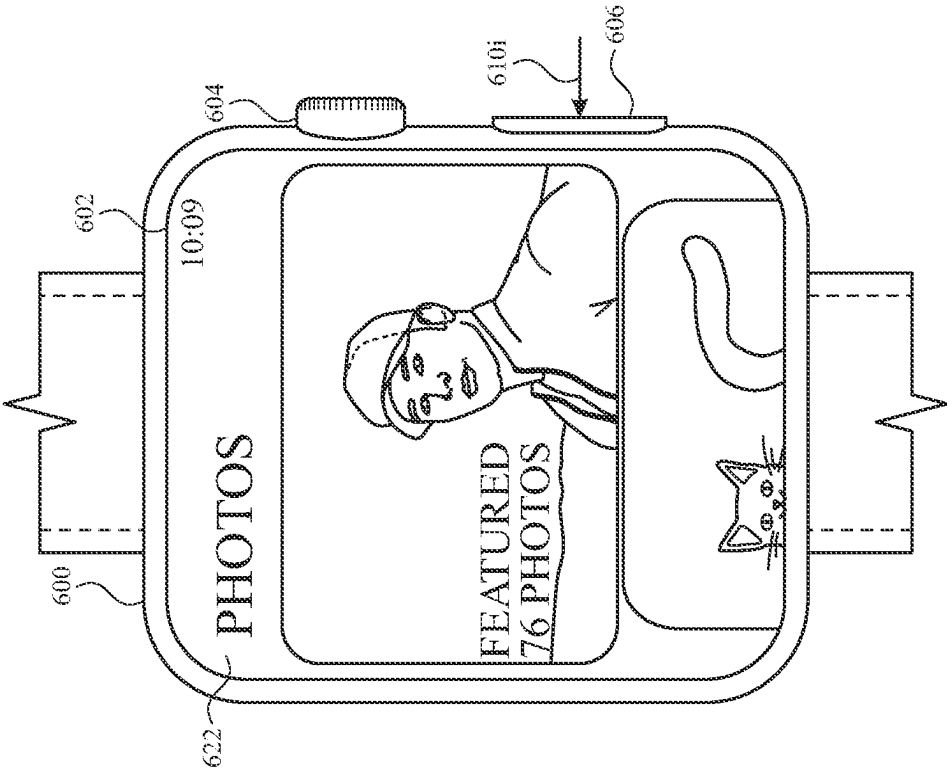


FIG. 6I

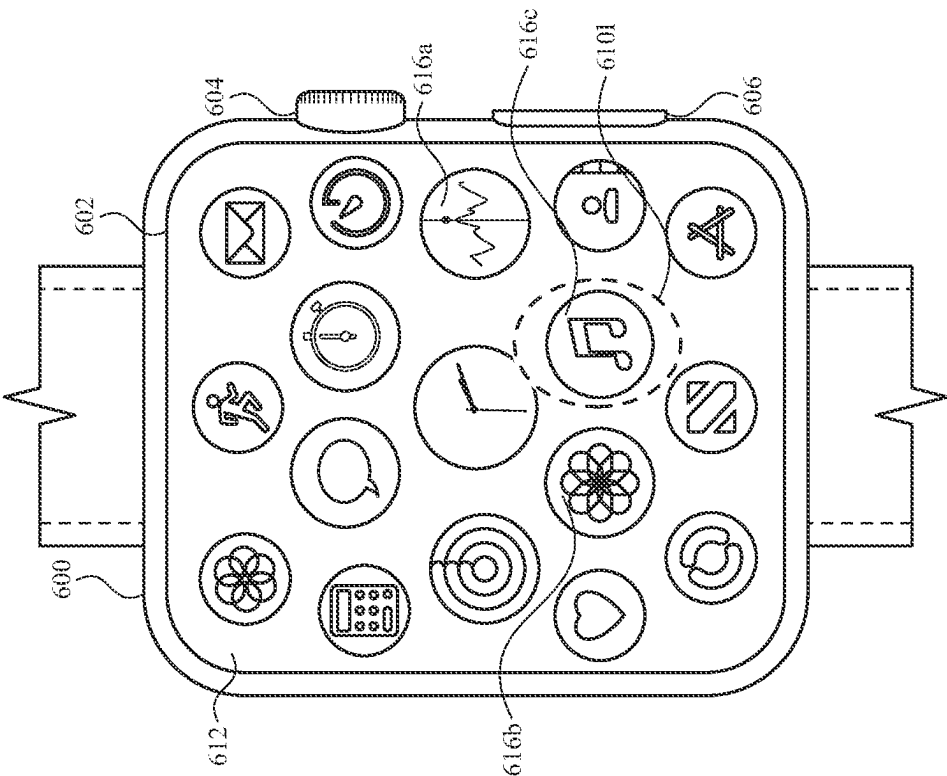


FIG. 6L

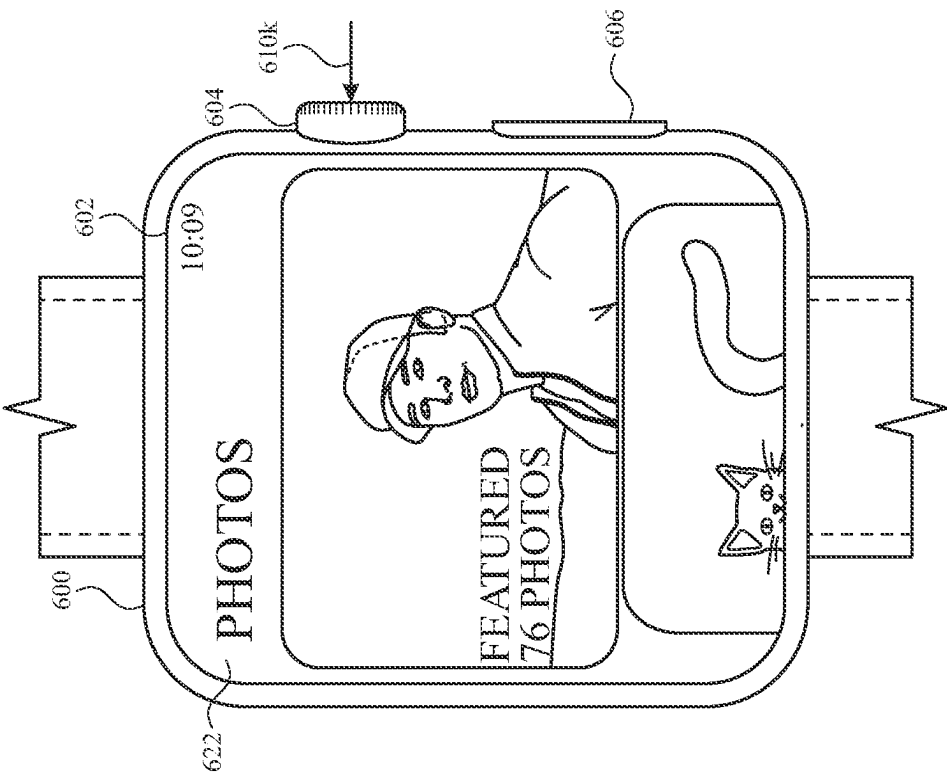
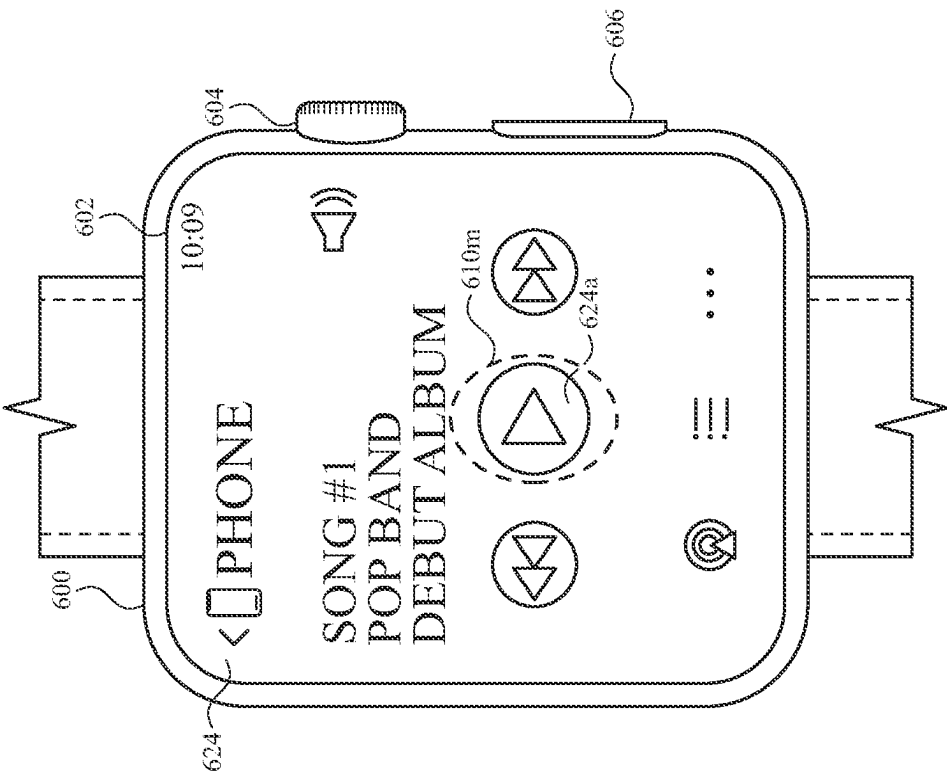
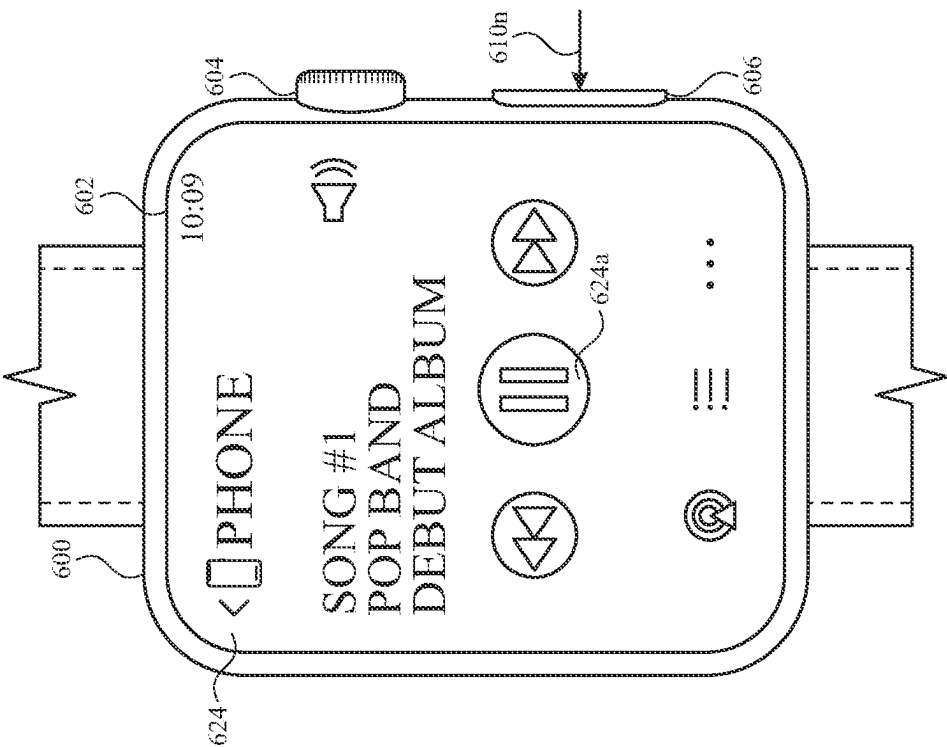


FIG. 6K



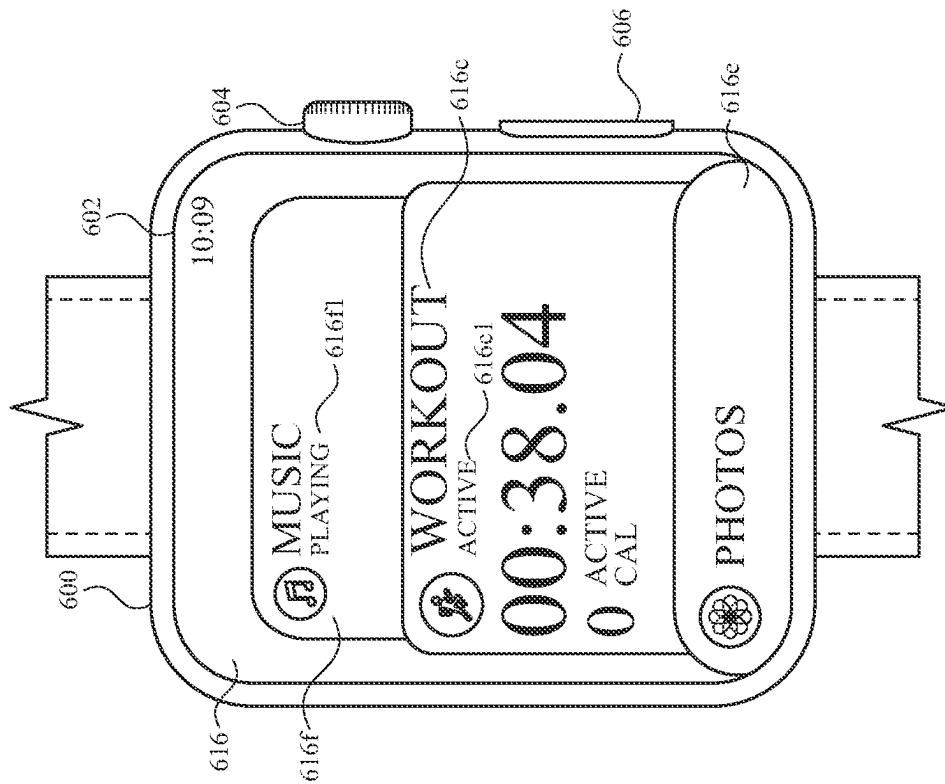
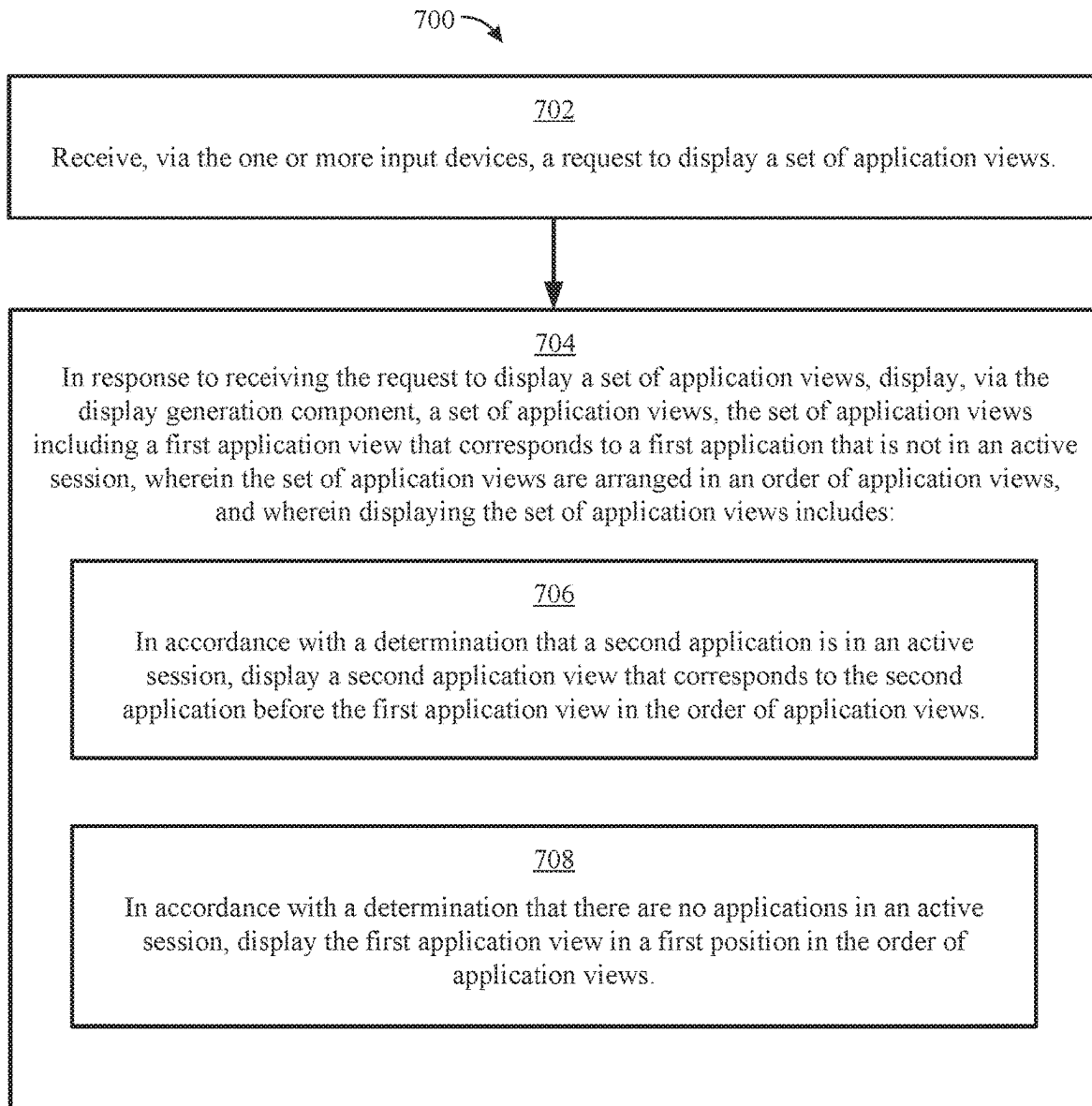


FIG. 60

**FIG. 7**

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DISPLAYING APPLICATION VIEWS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Patent Application No. 63/349,135, entitled “DISPLAYING APPLICATION VIEWS,” filed Jun. 5, 2022, the content of which is hereby incorporated by reference in its entirety.

FIELD

The present disclosure relates generally to computer user interfaces, and more specifically to techniques for displaying a set of application views.

BACKGROUND

Electronic devices can include applications for performing various functions. Electronic devices can also display user interfaces for accessing applications on the electronic device.

BRIEF SUMMARY

Some techniques for displaying a set of application views using electronic devices, however, are generally cumbersome and inefficient. For example, some existing techniques use a complex and time-consuming user interface, which may include multiple key presses or keystrokes. Existing techniques require more time than necessary, wasting user time and device energy. This latter consideration is particularly important in battery-operated devices.

Accordingly, the present technique provides electronic devices with faster, more efficient methods and interfaces for displaying a set of application views. Such methods and interfaces optionally complement or replace other methods for displaying a set of application views. Such methods and interfaces reduce the cognitive burden on a user and produce a more efficient human-machine interface. For battery-operated computing devices, such methods and interfaces conserve power and increase the time between battery charges.

In accordance with some embodiments, a method is described. The method comprises: at a computer system that is in communication with a display generation component and one or more input devices: receiving, via the one or more input devices, a request to display a set of application views; and in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

In accordance with some embodiments, a non-transitory computer-readable storage medium is described. The non-transitory computer-readable storage medium stores one or more programs configured to be executed by one or more

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processors of a computer system that is in communication with a display generation component and one or more input devices, the one or more programs including instructions for: receiving, via the one or more input devices, a request to display a set of application views; and in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

In accordance with some embodiments, a transitory computer-readable storage medium is described. The transitory computer-readable storage medium stores one or more programs configured to be executed by one or more processors of a computer system that is in communication with a display generation component and one or more input devices, the one or more programs including instructions for: receiving, via the one or more input devices, a request to display a set of application views; and in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

In accordance with some embodiments, a computer system configured to communicate with a display generation component and one or more input devices is described. The computer system comprises: one or more processors; and memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for: receiving, via the one or more input devices, a request to display a set of application views; and in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

In accordance with some embodiments, a computer system configured to communicate with a display generation

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component and one or more input devices is described. The computer system comprises: means for receiving, via the one or more input devices, a request to display a set of application views; and means for, in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

In accordance with some embodiments, a computer program product is described. The computer program product comprises one or more programs configured to be executed by one or more processors of a computer system that is in communication with a display generation component and one or more input devices, the one or more programs including instructions for: receiving, via the one or more input devices, a request to display a set of application views; and in response to receiving the request to display a set of application views, displaying, via the display generation component, a set of application views, the set of application views including a first application view that corresponds to a first application that is not in an active session, wherein the set of application views are arranged in an order of application views, and wherein displaying the set of application views includes: in accordance with a determination that a second application is in an active session, displaying a second application view that corresponds to the second application before the first application view in the order of application views; and in accordance with a determination that there are no applications in an active session, displaying the first application view in a first position in the order of application views.

Executable instructions for performing these functions are, optionally, included in a non-transitory computer-readable storage medium or other computer program product configured for execution by one or more processors. Executable instructions for performing these functions are, optionally, included in a transitory computer-readable storage medium or other computer program product configured for execution by one or more processors.

Thus, devices are provided with faster, more efficient methods and interfaces for displaying a set of application views, thereby increasing the effectiveness, efficiency, and user satisfaction with such devices. Such methods and interfaces may complement or replace other methods for displaying a set of application views.

DESCRIPTION OF THE FIGURES

For a better understanding of the various described embodiments, reference should be made to the Description of Embodiments below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

FIG. 1A is a block diagram illustrating a portable multifunction device with a touch-sensitive display in accordance with some embodiments.

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FIG. 1B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments.

FIG. 2 illustrates a portable multifunction device having a touch screen in accordance with some embodiments.

FIG. 3 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments.

FIG. 4A illustrates an exemplary user interface for a menu of applications on a portable multifunction device in accordance with some embodiments.

FIG. 4B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display in accordance with some embodiments.

FIG. 5A illustrates a personal electronic device in accordance with some embodiments.

FIG. 5B is a block diagram illustrating a personal electronic device in accordance with some embodiments.

FIGS. 6A-6O illustrate exemplary user interfaces for displaying a set of application views.

FIG. 7 is a flow diagram illustrating methods of displaying a set of application views in accordance with some embodiments.

DESCRIPTION OF EMBODIMENTS

The following description sets forth exemplary methods, parameters, and the like. It should be recognized, however, that such description is not intended as a limitation on the scope of the present disclosure but is instead provided as a description of exemplary embodiments.

There is a need for electronic devices that provide efficient methods and interfaces for displaying a set of application views. For example, a computer system can display a set of application views such that application views of applications that are in an active session are displayed before application views of application that are not in an active session. Such techniques can reduce the cognitive burden on a user who accesses applications via a set of application views, thereby enhancing productivity. Further, such techniques can reduce processor and battery power otherwise wasted on redundant user inputs.

Below, FIGS. 1A-1B, 2, 3, 4A-4B, and 5A-5B provide a description of exemplary devices for performing the techniques for displaying a set of application views. FIGS. 6A-6O illustrate exemplary user interfaces for displaying a set of application views. FIG. 7 is a flow diagram illustrating methods of displaying a set of application views in accordance with some embodiments. The user interfaces in FIGS. 6A-6O are used to illustrate the processes described below, including the processes in FIG. 7.

The processes described below enhance the operability of the devices and make the user-device interfaces more efficient (e.g., by helping the user to provide proper inputs and reducing user mistakes when operating/interacting with the device) through various techniques, including by providing improved visual feedback to the user, reducing the number of inputs needed to perform an operation, providing additional control options without cluttering the user interface with additional displayed controls, performing an operation when a set of conditions has been met without requiring further user input, and/or additional techniques. These techniques also reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

In addition, in methods described herein where one or more steps are contingent upon one or more conditions having been met, it should be understood that the described method can be repeated in multiple repetitions so that over the course of the repetitions all of the conditions upon which steps in the method are contingent have been met in different repetitions of the method. For example, if a method requires performing a first step if a condition is satisfied, and a second step if the condition is not satisfied, then a person of ordinary skill would appreciate that the claimed steps are repeated until the condition has been both satisfied and not satisfied, in no particular order. Thus, a method described with one or more steps that are contingent upon one or more conditions having been met could be rewritten as a method that is repeated until each of the conditions described in the method has been met. This, however, is not required of system or computer readable medium claims where the system or computer readable medium contains instructions for performing the contingent operations based on the satisfaction of the corresponding one or more conditions and thus is capable of determining whether the contingency has or has not been satisfied without explicitly repeating steps of a method until all of the conditions upon which steps in the method are contingent have been met. A person having ordinary skill in the art would also understand that, similar to a method with contingent steps, a system or computer readable storage medium can repeat the steps of a method as many times as are needed to ensure that all of the contingent steps have been performed.

Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. In some embodiments, these terms are used to distinguish one element from another. For example, a first touch could be termed a second touch, and, similarly, a second touch could be termed a first touch, without departing from the scope of the various described embodiments. In some embodiments, the first touch and the second touch are two separate references to the same touch. In some embodiments, the first touch and the second touch are both touches, but they are not the same touch.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

Embodiments of electronic devices, user interfaces for such devices, and associated processes for using such devices are described. In some embodiments, the device is a portable communications device, such as a mobile telephone, that also contains other functions, such as PDA and/or music player functions. Exemplary embodiments of portable multifunction devices include, without limitation, the iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other portable electronic devices, such as laptops or tablet computers with touch-sensitive surfaces (e.g., touch screen displays and/or touchpads), are, optionally, used. It should also be understood that, in some embodiments, the device is not a portable communications device, but is a desktop computer with a touch-sensitive surface (e.g., a touch screen display and/or a touchpad). In some embodiments, the electronic device is a computer system that is in communication (e.g., via wireless communication, via wired communication) with a display generation component. The display generation component is configured to provide visual output, such as display via a CRT display, display via an LED display, or display via image projection. In some embodiments, the display generation component is integrated with the computer system. In some embodiments, the display generation component is separate from the computer system. As used herein, “displaying” content includes causing to display the content (e.g., video data rendered or decoded by display controller 156) by transmitting, via a wired or wireless connection, data (e.g., image data or video data) to an integrated or external display generation component to visually produce the content.

In the discussion that follows, an electronic device that includes a display and a touch-sensitive surface is described. It should be understood, however, that the electronic device optionally includes one or more other physical user-interface devices, such as a physical keyboard, a mouse, and/or a joystick.

The device typically supports a variety of applications, such as one or more of the following: a drawing application, a presentation application, a word processing application, a website creation application, a disk authoring application, a spreadsheet application, a gaming application, a telephone application, a video conferencing application, an e-mail application, an instant messaging application, a workout support application, a photo management application, a digital camera application, a digital video camera application, a web browsing application, a digital music player application, and/or a digital video player application.

The various applications that are executed on the device optionally use at least one common physical user-interface device, such as the touch-sensitive surface. One or more functions of the touch-sensitive surface as well as corresponding information displayed on the device are, optionally, adjusted and/or varied from one application to the next and/or within a respective application. In this way, a common physical architecture (such as the touch-sensitive surface) of the device optionally supports the variety of applications with user interfaces that are intuitive and transparent to the user.

Attention is now directed toward embodiments of portable devices with touch-sensitive displays. FIG. 1A is a block diagram illustrating portable multifunction device 100 with touch-sensitive display system 112 in accordance with some embodiments. Touch-sensitive display 112 is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.” Device 100 includes memory 102 (which optionally

includes one or more computer-readable storage mediums), memory controller **122**, one or more processing units (CPUs) **120**, peripherals interface **118**, RF circuitry **108**, audio circuitry **110**, speaker **111**, microphone **113**, input/output (I/O) subsystem **106**, other input control devices **116**, and external port **124**. Device **100** optionally includes one or more optical sensors **164**. Device **100** optionally includes one or more contact intensity sensors **165** for detecting intensity of contacts on device **100** (e.g., a touch-sensitive surface such as touch-sensitive display system **112** of device **100**). Device **100** optionally includes one or more tactile output generators **167** for generating tactile outputs on device **100** (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **112** of device **100** or touchpad **355** of device **300**). These components optionally communicate over one or more communication buses or signal lines **103**.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the

user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device **100** is only one example of a portable multifunction device, and that device **100** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. **1A** are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory **102** optionally includes high-speed random access memory and optionally also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller **122** optionally controls access to memory **102** by other components of device **100**.

Peripherals interface **118** can be used to couple input and output peripherals of the device to CPU **120** and memory **102**. The one or more processors **120** run or execute various software programs (such as computer programs (e.g., including instructions)) and/or sets of instructions stored in memory **102** to perform various functions for device **100** and to process data. In some embodiments, peripherals interface **118**, CPU **120**, and memory controller **122** are, optionally, implemented on a single chip, such as chip **104**. In some other embodiments, they are, optionally, implemented on separate chips.

RF (radio frequency) circuitry **108** receives and sends RF signals, also called electromagnetic signals. RF circuitry **108** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **108** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry **108** optionally communicates with networks, such as the Internet, also referred to

as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **108** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPDA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, a protocol for e-mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **110**, speaker **111**, and microphone **113** provide an audio interface between a user and device **100**. Audio circuitry **110** receives audio data from peripherals interface **118**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **111**. Speaker **111** converts the electrical signal to human-audible sound waves. Audio circuitry **110** also receives electrical signals converted by microphone **113** from sound waves. Audio circuitry **110** converts the electrical signal to audio data and transmits the audio data to peripherals interface **118** for processing. Audio data is, optionally, retrieved from and/or transmitted to memory **102** and/or RF circuitry **108** by peripherals interface **118**. In some embodiments, audio circuitry **110** also includes a headset jack (e.g., **212**, FIG. 2). The headset jack provides an interface between audio circuitry **110** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **106** couples input/output peripherals on device **100**, such as touch screen **112** and other input control devices **116**, to peripherals interface **118**. I/O subsystem **106** optionally includes display controller **156**, optical sensor controller **158**, depth camera controller **169**, intensity sensor controller **159**, haptic feedback controller **161**, and one or more input controllers **160** for other input or control devices. The one or more input controllers **160** receive/send electrical signals from/to other input control devices **116**. The other input control devices **116** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some embodiments, input controller(s) **160** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **208**, FIG. 2) optionally include an up/down button for volume control of speaker **111** and/or microphone **113**. The one or more

buttons optionally include a push button (e.g., **206**, FIG. 2). In some embodiments, the electronic device is a computer system that is in communication (e.g., via wireless communication, via wired communication) with one or more input devices. In some embodiments, the one or more input devices include a touch-sensitive surface (e.g., a trackpad, as part of a touch-sensitive display). In some embodiments, the one or more input devices include one or more camera sensors (e.g., one or more optical sensors **164** and/or one or more depth camera sensors **175**), such as for tracking a user's gestures (e.g., hand gestures and/or air gestures) as input. In some embodiments, the one or more input devices are integrated with the computer system. In some embodiments, the one or more input devices are separate from the computer system. In some embodiments, an air gesture is a gesture that is detected without the user touching an input element that is part of the device (or independently of an input element that is a part of the device) and is based on detected motion of a portion of the user's body through the air including motion of the user's body relative to an absolute reference (e.g., an angle of the user's arm relative to the ground or a distance of the user's hand relative to the ground), relative to another portion of the user's body (e.g., movement of a hand of the user relative to a shoulder of the user, movement of one hand of the user relative to another hand of the user, and/or movement of a finger of the user relative to another finger or portion of a hand of the user), and/or absolute motion of a portion of the user's body (e.g., a tap gesture that includes movement of a hand in a predetermined pose by a predetermined amount and/or speed, or a shake gesture that includes a predetermined speed or amount of rotation of a portion of the user's body).

A quick press of the push button optionally disengages a lock of touch screen **112** or optionally begins a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **206**) optionally turns power to device **100** on or off. The functionality of one or more of the buttons are, optionally, user-customizable. Touch screen **112** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **112** provides an input interface and an output interface between the device and a user. Display controller **156** receives and/or sends electrical signals from/to touch screen **112**. Touch screen **112** displays visual output to the user. The visual output optionally includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output optionally corresponds to user-interface objects.

Touch screen **112** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **112** and display controller **156** (along with any associated modules and/or sets of instructions in memory **102**) detect contact (and any movement or breaking of the contact) on touch screen **112** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **112**. In an exemplary embodiment, a point of contact between touch screen **112** and the user corresponds to a finger of the user.

Touch screen **112** optionally uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology,

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although other display technologies are used in other embodiments. Touch screen **112** and display controller **156** optionally detect contact and any movement or breaking thereof using any of a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **112**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **112** is, optionally, analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **112** displays visual output from device **100**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **112** is described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **112** optionally has a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user optionally makes contact with touch screen **112** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **100** optionally includes a touchpad for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is, optionally, a touch-sensitive surface that is separate from touch screen **112** or an extension of the touch-sensitive surface formed by the touch screen.

Device **100** also includes power system **162** for powering the various components. Power system **162** optionally

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includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode (LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device **100** optionally also includes one or more optical sensors **164**. FIG. 1A shows an optical sensor coupled to optical sensor controller **158** in I/O subsystem **106**. Optical sensor **164** optionally includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor **164** receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module **143** (also called a camera module), optical sensor **164** optionally captures still images or video. In some embodiments, an optical sensor is located on the back of device **100**, opposite touch screen display **112** on the front of the device so that the touch screen display is enabled for use as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor **164** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor **164** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **100** optionally also includes one or more depth camera sensors **175**. FIG. 1A shows a depth camera sensor coupled to depth camera controller **169** in I/O subsystem **106**. Depth camera sensor **175** receives data from the environment to create a three dimensional model of an object (e.g., a face) within a scene from a viewpoint (e.g., a depth camera sensor). In some embodiments, in conjunction with imaging module **143** (also called a camera module), depth camera sensor **175** is optionally used to determine a depth map of different portions of an image captured by the imaging module **143**. In some embodiments, a depth camera sensor is located on the front of device **100** so that the user's image with depth information is, optionally, obtained for video conferencing while the user views the other video conference participants on the touch screen display and to capture selfies with depth map data. In some embodiments, the depth camera sensor **175** is located on the back of device, or on the back and the front of the device **100**. In some embodiments, the position of depth camera sensor **175** can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a depth camera sensor **175** is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device **100** optionally also includes one or more contact intensity sensors **165**. FIG. 1A shows a contact intensity sensor coupled to intensity sensor controller **159** in I/O subsystem **106**. Contact intensity sensor **165** optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor **165** receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is

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collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 112). In some embodiments, at least one contact intensity sensor is located on the back of device 100, opposite touch screen display 112, which is located on the front of device 100.

Device 100 optionally also includes one or more proximity sensors 166. FIG. 1A shows proximity sensor 166 coupled to peripherals interface 118. Alternately, proximity sensor 166 is, optionally, coupled to input controller 160 in I/O subsystem 106. Proximity sensor 166 optionally performs as described in U.S. patent application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen 112 when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device 100 optionally also includes one or more tactile output generators 167. FIG. 1A shows a tactile output generator coupled to haptic feedback controller 161 in I/O subsystem 106. Tactile output generator 167 optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor 165 receives tactile feedback generation instructions from haptic feedback module 133 and generates tactile outputs on device 100 that are capable of being sensed by a user of device 100. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 112) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device 100) or laterally (e.g., back and forth in the same plane as a surface of device 100). In some embodiments, at least one tactile output generator sensor is located on the back of device 100, opposite touch screen display 112, which is located on the front of device 100.

Device 100 optionally also includes one or more accelerometers 168. FIG. 1A shows accelerometer 168 coupled to peripherals interface 118. Alternately, accelerometer 168 is, optionally, coupled to an input controller 160 in I/O subsystem 106. Accelerometer 168 optionally performs as described in U.S. Patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. Patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device 100 optionally includes, in addition to accelerometer(s) 168, a magnetometer and a GPS (or GLONASS or other global navigation system) receiver for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device 100.

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In some embodiments, the software components stored in memory 102 include operating system 126, communication module (or set of instructions) 128, contact/motion module (or set of instructions) 130, graphics module (or set of instructions) 132, text input module (or set of instructions) 134, Global Positioning System (GPS) module (or set of instructions) 135, and applications (or sets of instructions) 136. Furthermore, in some embodiments, memory 102 (FIG. 1A) or 370 (FIG. 3) stores device/global internal state 157, as shown in FIGS. 1A and 3. Device/global internal state 157 includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display 112; sensor state, including information obtained from the device's various sensors and input control devices 116; and location information concerning the device's location and/or attitude.

Operating system 126 (e.g., Darwin, RTXC, LINUX, UNIX, OS X, iOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module 128 facilitates communication with other devices over one or more external ports 124 and also includes various software components for handling data received by RF circuitry 108 and/or external port 124. External port 124 (e.g., Universal Serial Bus (USB), FIREWIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module 130 optionally detects contact with touch screen 112 (in conjunction with display controller 156) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module 130 includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module 130 receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., "multitouch"/multiple finger contacts). In some embodiments, contact/motion module 130 and display controller 156 detect contact on a touchpad.

In some embodiments, contact/motion module 130 uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has "clicked" on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation

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thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device 100. For example, a mouse “click” threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click “intensity” parameter).

Contact/motion module 130 optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module 132 includes various known software components for rendering and displaying graphics on touch screen 112 or other display, including components for changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.

In some embodiments, graphics module 132 stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module 132 receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller 156.

Haptic feedback module 133 includes various software components for generating instructions used by tactile output generator(s) 167 to produce tactile outputs at one or more locations on device 100 in response to user interactions with device 100.

Text input module 134, which is, optionally, a component of graphics module 132, provides soft keyboards for entering text in various applications (e.g., contacts 137, e-mail client module 140, IM module 141, browser module 147, and any other application that needs text input).

GPS module 135 determines the location of the device and provides this information for use in various applications (e.g., to telephone module 138 for use in location-based dialing; to camera module 143 as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Applications 136 optionally include the following modules (or sets of instructions), or a subset or superset thereof:

Contacts module 137 (sometimes called an address book or contact list);

Telephone module 138;

Video conference module 139;

E-mail client module 140;

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Instant messaging (IM) module 141;

Workout support module 142;

Camera module 143 for still and/or video images;

Image management module 144;

Video player module;

Music player module;

Browser module 147;

Calendar module 148;

Widget modules 149, which optionally include one or more of: weather widget 149-1, stocks widget 149-2, calculator widget 149-3, alarm clock widget 149-4, dictionary widget 149-5, and other widgets obtained by the user, as well as user-created widgets 149-6;

Widget creator module 150 for making user-created widgets 149-6;

Search module 151;

Video and music player module 152, which merges video player module and music player module;

Notes module 153;

Map module 154; and/or

Online video module 155.

Examples of other applications 136 that are, optionally, stored in memory 102 include other word processing applications, other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, contacts module 137 are, optionally, used to manage an address book or contact list (e.g., stored in application internal state 192 of contacts module 137 in memory 102 or memory 370), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone module 138, video conference module 139, e-mail client module 140, or IM module 141; and so forth.

In conjunction with RF circuitry 108, audio circuitry 110, speaker 111, microphone 113, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, telephone module 138 are optionally, used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module 137, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry 108, audio circuitry 110, speaker 111, microphone 113, touch screen 112, display controller 156, optical sensor 164, optical sensor controller 158, contact/motion module 130, graphics module 132, text input module 134, contacts module 137, and telephone module 138, video conference module 139 includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry 108, touch screen 112, display controller 156, contact/motion module 130, graphics module 132, and text input module 134, e-mail client module 140 includes executable instructions to create, send, receive, and manage e-mail in response to user instructions.

In conjunction with image management module **144**, e-mail client module **140** makes it very easy to create and send e-mails with still or video images taken with camera module **143**.

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, the instant messaging module **141** includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages optionally include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, "instant messaging" refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, GPS module **135**, map module **154**, and music player module, workout support module **142** includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen **112**, display controller **156**, optical sensor(s) **164**, optical sensor controller **158**, contact/motion module **130**, graphics module **132**, and image management module **144**, camera module **143** includes executable instructions to capture still images or video (including a video stream) and store them into memory **102**, modify characteristics of a still image or video, or delete a still image or video from memory **102**.

In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and camera module **143**, image management module **144** includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, browser module **147** includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, e-mail client module **140**, and browser module **147**, calendar module **148** includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and browser module **147**, widget modules **149** are mini-applications that are,

optionally, downloaded and used by a user (e.g., weather widget **149-1**, stocks widget **149-2**, calculator widget **149-3**, alarm clock widget **149-4**, and dictionary widget **149-5**) or created by the user (e.g., user-created widget **149-6**). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, and browser module **147**, the widget creator module **150** are, optionally, used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, search module **151** includes executable instructions to search for text, music, sound, image, video, and/or other files in memory **102** that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, audio circuitry **110**, speaker **111**, RF circuitry **108**, and browser module **147**, video and music player module **152** includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen **112** or on an external, connected display via external port **124**). In some embodiments, device **100** optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, and text input module **134**, notes module **153** includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry **108**, touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, text input module **134**, GPS module **135**, and browser module **147**, map module **154** are, optionally, used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen **112**, display controller **156**, contact/motion module **130**, graphics module **132**, audio circuitry **110**, speaker **111**, RF circuitry **108**, text input module **134**, e-mail client module **140**, and browser module **147**, online video module **155** includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port **124**), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module **141**, rather than e-mail client module **140**, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for

Playing Online Videos,” filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs (such as computer programs (e.g., including instructions)), procedures, or modules, and thus various subsets of these modules are, optionally, combined or otherwise rearranged in various embodiments. For example, video player module is, optionally, combined with music player module into a single module (e.g., video and music player module **152**, FIG. 1A). In some embodiments, memory **102** optionally stores a subset of the modules and data structures identified above. Furthermore, memory **102** optionally stores additional modules and data structures not described above.

In some embodiments, device **100** is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device **100**, the number of physical input control devices (such as push buttons, dials, and the like) on device **100** is, optionally, reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodiments, the touchpad, when touched by the user, navigates device **100** to a main, home, or root menu from any user interface that is displayed on device **100**. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. 1B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory **102** (FIG. 1A) or **370** (FIG. 3) includes event sorter **170** (e.g., in operating system **126**) and a respective application **136-1** (e.g., any of the aforementioned applications **137-151**, **155**, **380-390**).

Event sorter **170** receives event information and determines the application **136-1** and application view **191** of application **136-1** to which to deliver the event information. Event sorter **170** includes event monitor **171** and event dispatcher module **174**. In some embodiments, application **136-1** includes application internal state **192**, which indicates the current application view(s) displayed on touch-sensitive display **112** when the application is active or executing. In some embodiments, device/global internal state **157** is used by event sorter **170** to determine which application(s) is (are) currently active, and application internal state **192** is used by event sorter **170** to determine application views **191** to which to deliver event information.

In some embodiments, application internal state **192** includes additional information, such as one or more of: resume information to be used when application **136-1** resumes execution, user interface state information that indicates information being displayed or that is ready for display by application **136-1**, a state queue for enabling the user to go back to a prior state or view of application **136-1**, and a redo/undo queue of previous actions taken by the user.

Event monitor **171** receives event information from peripherals interface **118**. Event information includes infor-

mation about a sub-event (e.g., a user touch on touch-sensitive display **112**, as part of a multi-touch gesture). Peripherals interface **118** transmits information it receives from I/O subsystem **106** or a sensor, such as proximity sensor **166**, accelerometer(s) **168**, and/or microphone **113** (through audio circuitry **110**). Information that peripherals interface **118** receives from I/O subsystem **106** includes information from touch-sensitive display **112** or a touch-sensitive surface.

In some embodiments, event monitor **171** sends requests to the peripherals interface **118** at predetermined intervals. In response, peripherals interface **118** transmits event information. In other embodiments, peripherals interface **118** transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter **170** also includes a hit view determination module **172** and/or an active event recognizer determination module **173**.

Hit view determination module **172** provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display **112** displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which a touch is detected optionally correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is, optionally, called the hit view, and the set of events that are recognized as proper inputs are, optionally, determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module **172** receives information related to sub-events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module **172** identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module **172**, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module **173** determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module **173** determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module **173** determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module **174** dispatches the event information to an event recognizer (e.g., event recognizer **180**). In embodiments including active event recognizer determination module **173**, event dispatcher module **174** delivers

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the event information to an event recognizer determined by active event recognizer determination module 173. In some embodiments, event dispatcher module 174 stores in an event queue the event information, which is retrieved by a respective event receiver 182.

In some embodiments, operating system 126 includes event sorter 170. Alternatively, application 136-1 includes event sorter 170. In yet other embodiments, event sorter 170 is a stand-alone module, or a part of another module stored in memory 102, such as contact/motion module 130.

In some embodiments, application 136-1 includes a plurality of event handlers 190 and one or more application views 191, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view 191 of the application 136-1 includes one or more event recognizers 180. Typically, a respective application view 191 includes a plurality of event recognizers 180. In other embodiments, one or more of event recognizers 180 are part of a separate module, such as a user interface kit or a higher level object from which application 136-1 inherits methods and other properties. In some embodiments, a respective event handler 190 includes one or more of: data updater 176, object updater 177, GUI updater 178, and/or event data 179 received from event sorter 170. Event handler 190 optionally utilizes or calls data updater 176, object updater 177, or GUI updater 178 to update the application internal state 192. Alternatively, one or more of the application views 191 include one or more respective event handlers 190. Also, in some embodiments, one or more of data updater 176, object updater 177, and GUI updater 178 are included in a respective application view 191.

A respective event recognizer 180 receives event information (e.g., event data 179) from event sorter 170 and identifies an event from the event information. Event recognizer 180 includes event receiver 182 and event comparator 184. In some embodiments, event recognizer 180 also includes at least a subset of: metadata 183, and event delivery instructions 188 (which optionally include sub-event delivery instructions).

Event receiver 182 receives event information from event sorter 170. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information optionally also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator 184 compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub-event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator 184 includes event definitions 186. Event definitions 186 contain definitions of events (e.g., predefined sequences of sub-events), for example, event 1 (187-1), event 2 (187-2), and others. In some embodiments, sub-events in an event (e.g., 187-1 and/or 187-2) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event 1 (187-1) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a

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predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event 2 (187-2) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display 112, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers 190.

In some embodiments, event definitions 186 include a definition of an event for a respective user-interface object. In some embodiments, event comparator 184 performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display 112, when a touch is detected on touch-sensitive display 112, event comparator 184 performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler 190, the event comparator uses the result of the hit test to determine which event handler 190 should be activated. For example, event comparator 184 selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (187) also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer 180 determines that the series of sub-events do not match any of the events in event definitions 186, the respective event recognizer 180 enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer 180 includes metadata 183 with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata 183 includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata 183 includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer 180 activates event handler 190 associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer 180 delivers event information associated with the event to event handler 190. Activating an event handler 190 is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer 180 throws a flag associated with the recognized event, and event handler 190 associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions 188 include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the

series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **176** creates and updates data used in application **136-1**. For example, data updater **176** updates the telephone number used in contacts module **137**, or stores a video file used in video player module. In some embodiments, object updater **177** creates and updates objects used in application **136-1**. For example, object updater **177** creates a new user-interface object or updates the position of a user-interface object. GUI updater **178** updates the GUI. For example, GUI updater **178** prepares display information and sends it to graphics module **132** for display on a touch-sensitive display.

In some embodiments, event handler(s) **190** includes or has access to data updater **176**, object updater **177**, and GUI updater **178**. In some embodiments, data updater **176**, object updater **177**, and GUI updater **178** are included in a single module of a respective application **136-1** or application view **191**. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive displays also applies to other forms of user inputs to operate multifunction devices **100** with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. **2** illustrates a portable multifunction device **100** having a touch screen **112** in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) **200**. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers **202** (not drawn to scale in the figure) or one or more styluses **203** (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device **100**. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device **100** optionally also include one or more physical buttons, such as "home" or menu button **204**. As described previously, menu button **204** is, optionally, used to navigate to any application **136** in a set of applications that are, optionally, executed on device **100**. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen **112**.

In some embodiments, device **100** includes touch screen **112**, menu button **204**, push button **206** for powering the device on/off and locking the device, volume adjustment button(s) **208**, subscriber identity module (SIM) card slot **210**, headset jack **212**, and docking/charging external port

124. Push button **206** is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device **100** also accepts verbal input for activation or deactivation of some functions through microphone **113**. Device **100** also, optionally, includes one or more contact intensity sensors **165** for detecting intensity of contacts on touch screen **112** and/or one or more tactile output generators **167** for generating tactile outputs for a user of device **100**.

FIG. **3** is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device **300** need not be portable. In some embodiments, device **300** is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child's learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device **300** typically includes one or more processing units (CPUs) **310**, one or more network or other communications interfaces **360**, memory **370**, and one or more communication buses **320** for interconnecting these components. Communication buses **320** optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device **300** includes input/output (I/O) interface **330** comprising display **340**, which is typically a touch screen display. I/O interface **330** also optionally includes a keyboard and/or mouse (or other pointing device) **350** and touchpad **355**, tactile output generator **357** for generating tactile outputs on device **300** (e.g., similar to tactile output generator(s) **167** described above with reference to FIG. **1A**), sensors **359** (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) **165** described above with reference to FIG. **1A**). Memory **370** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory **370** optionally includes one or more storage devices remotely located from CPU(s) **310**. In some embodiments, memory **370** stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory **102** of portable multifunction device **100** (FIG. **1A**), or a subset thereof. Furthermore, memory **370** optionally stores additional programs, modules, and data structures not present in memory **102** of portable multifunction device **100**. For example, memory **370** of device **300** optionally stores drawing module **380**, presentation module **382**, word processing module **384**, website creation module **386**, disk authoring module **388**, and/or spreadsheet module **390**, while memory **102** of portable multifunction device **100** (FIG. **1A**) optionally does not store these modules.

Each of the above-identified elements in FIG. **3** is, optionally, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or computer programs (e.g., sets of instructions or including instructions) need not be implemented as separate software programs (such as computer programs (e.g., including instructions)), procedures, or modules, and thus various subsets of these

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modules are, optionally, combined or otherwise rearranged in various embodiments. In some embodiments, memory 370 optionally stores a subset of the modules and data structures identified above. Furthermore, memory 370 optionally stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that are, optionally, implemented on, for example, portable multifunction device 100.

FIG. 4A illustrates an exemplary user interface for a menu of applications on portable multifunction device 100 in accordance with some embodiments. Similar user interfaces are, optionally, implemented on device 300. In some embodiments, user interface 400 includes the following elements, or a subset or superset thereof:

Signal strength indicator(s) 402 for wireless communication(s), such as cellular and Wi-Fi signals;

Time 404;

Bluetooth indicator 405;

Battery status indicator 406;

Tray 408 with icons for frequently used applications, such as:

Icon 416 for telephone module 138, labeled "Phone," which optionally includes an indicator 414 of the number of missed calls or voicemail messages;

Icon 418 for e-mail client module 140, labeled "Mail," which optionally includes an indicator 410 of the number of unread e-mails;

Icon 420 for browser module 147, labeled "Browser;" and

Icon 422 for video and music player module 152, also referred to as iPod (trademark of Apple Inc.) module 152, labeled "iPod;" and

Icons for other applications, such as:

Icon 424 for IM module 141, labeled "Messages;"

Icon 426 for calendar module 148, labeled "Calendar;"

Icon 428 for image management module 144, labeled "Photos;"

Icon 430 for camera module 143, labeled "Camera;"

Icon 432 for online video module 155, labeled "Online Video;"

Icon 434 for stocks widget 149-2, labeled "Stocks;"

Icon 436 for map module 154, labeled "Maps;"

Icon 438 for weather widget 149-1, labeled "Weather;"

Icon 440 for alarm clock widget 149-4, labeled "Clock;"

Icon 442 for workout support module 142, labeled "Workout Support;"

Icon 444 for notes module 153, labeled "Notes;" and

Icon 446 for a settings application or module, labeled "Settings," which provides access to settings for device 100 and its various applications 136.

It should be noted that the icon labels illustrated in FIG. 4A are merely exemplary. For example, icon 422 for video and music player module 152 is labeled "Music" or "Music Player." Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. 4B illustrates an exemplary user interface on a device (e.g., device 300, FIG. 3) with a touch-sensitive surface 451 (e.g., a tablet or touchpad 355, FIG. 3) that is separate from the display 450 (e.g., touch screen display 112). Device 300 also, optionally, includes one or more

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contact intensity sensors (e.g., one or more of sensors 359) for detecting intensity of contacts on touch-sensitive surface 451 and/or one or more tactile output generators 357 for generating tactile outputs for a user of device 300.

Although some of the examples that follow will be given with reference to inputs on touch screen display 112 (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. 4B. In some embodiments, the touch-sensitive surface (e.g., 451 in FIG. 4B) has a primary axis (e.g., 452 in FIG. 4B) that corresponds to a primary axis (e.g., 453 in FIG. 4B) on the display (e.g., 450). In accordance with these embodiments, the device detects contacts (e.g., 460 and 462 in FIG. 4B) with the touch-sensitive surface 451 at locations that correspond to respective locations on the display (e.g., in FIG. 4B, 460 corresponds to 468 and 462 corresponds to 470). In this way, user inputs (e.g., contacts 460 and 462, and movements thereof) detected by the device on the touch-sensitive surface (e.g., 451 in FIG. 4B) are used by the device to manipulate the user interface on the display (e.g., 450 in FIG. 4B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. 5A illustrates exemplary personal electronic device 500. Device 500 includes body 502. In some embodiments, device 500 can include some or all of the features described with respect to devices 100 and 300 (e.g., FIGS. 1A-4B). In some embodiments, device 500 has touch-sensitive display screen 504, hereafter touch screen 504. Alternatively, or in addition to touch screen 504, device 500 has a display and a touch-sensitive surface. As with devices 100 and 300, in some embodiments, touch screen 504 (or the touch-sensitive surface) optionally includes one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen 504 (or the touch-sensitive surface) can provide output data that represents the intensity of touches. The user interface of device 500 can respond to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device 500.

Exemplary techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled "Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application," filed May 8, 2013, published as WIPO Publication No. WO/2013/169849, and International Patent

Application Serial No. PCT/US2013/069483, titled “Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships,” filed Nov. 11, 2013, published as WIPO Publication No. WO/2014/105276, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **500** has one or more input mechanisms **506** and **508**. Input mechanisms **506** and **508**, if included, can be physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **500** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **500** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **500** to be worn by a user.

FIG. **5B** depicts exemplary personal electronic device **500**. In some embodiments, device **500** can include some or all of the components described with respect to FIGS. **1A**, **1B**, and **3**. Device **500** has bus **512** that operatively couples I/O section **514** with one or more computer processors **516** and memory **518**. I/O section **514** can be connected to display **504**, which can have touch-sensitive component **522** and, optionally, intensity sensor **524** (e.g., contact intensity sensor). In addition, I/O section **514** can be connected with communication unit **530** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device **500** can include input mechanisms **506** and/or **508**. Input mechanism **506** is, optionally, a rotatable input device or a depressible and rotatable input device, for example. Input mechanism **508** is, optionally, a button, in some examples.

Input mechanism **508** is, optionally, a microphone, in some examples. Personal electronic device **500** optionally includes various sensors, such as GPS sensor **532**, accelerometer **534**, directional sensor **540** (e.g., compass), gyroscope **536**, motion sensor **538**, and/or a combination thereof, all of which can be operatively connected to I/O section **514**.

Memory **518** of personal electronic device **500** can include one or more non-transitory computer-readable storage mediums, for storing computer-executable instructions, which, when executed by one or more computer processors **516**, for example, can cause the computer processors to perform the techniques described below, including process **700** (FIG. **7**). A computer-readable storage medium can be any medium that can tangibly contain or store computer-executable instructions for use by or in connection with the instruction execution system, apparatus, or device. In some examples, the storage medium is a transitory computer-readable storage medium. In some examples, the storage medium is a non-transitory computer-readable storage medium. The non-transitory computer-readable storage medium can include, but is not limited to, magnetic, optical, and/or semiconductor storages. Examples of such storage include magnetic disks, optical discs based on CD, DVD, or Blu-ray technologies, as well as persistent solid-state memory such as flash, solid-state drives, and the like. Personal electronic device **500** is not limited to the components and configuration of FIG. **5B**, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, optionally, displayed on the display screen of devices **100**, **300**, and/or

500 (FIGS. **1A**, **3**, and **5A-5B**). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each optionally constitute an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad **355** in FIG. **3** or touch-sensitive surface **451** in FIG. **4B**) while the cursor is over a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system **112** in FIG. **1A** or touch screen **112** in FIG. **4A**) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally, based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when

the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds optionally includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation), rather than being used to determine whether to perform a first operation or a second operation.

As used herein, an “installed application” refers to a software application that has been downloaded onto an electronic device (e.g., devices 100, 300, and/or 500) and is ready to be launched (e.g., become opened) on the device. In some embodiments, a downloaded application becomes an installed application by way of an installation program that extracts program portions from a downloaded package and integrates the extracted portions with the operating system of the computer system.

As used herein, the terms “open application” or “executing application” refer to a software application with retained state information (e.g., as part of device/global internal state 157 and/or application internal state 192). An open or executing application is, optionally, any one of the following types of applications:

- an active application, which is currently displayed on a display screen of the device that the application is being used on;
- a background application (or background processes), which is not currently displayed, but one or more processes for the application are being processed by one or more processors; and
- a suspended or hibernated application, which is not running, but has state information that is stored in memory (volatile and non-volatile, respectively) and that can be used to resume execution of the application.

As used herein, the term “closed application” refers to software applications without retained state information (e.g., state information for closed applications is not stored in a memory of the device). Accordingly, closing an application includes stopping and/or removing application processes for the application and removing state information for the application from the memory of the device. Generally, opening a second application while in a first application does not close the first application. When the second application is displayed and the first application ceases to be displayed, the first application becomes a background application.

Attention is now directed towards embodiments of user interfaces (“UP”) and associated processes that are implemented on an electronic device, such as portable multifunction device 100, device 300, or device 500.

FIGS. 6A-6O illustrate exemplary user interfaces for displaying a set of application views, in accordance with some embodiments. The user interfaces in these figures are used to illustrate the processes described below, including the processes in FIG. 7.

FIG. 6A illustrates computer system 600, which includes display 602, input mechanism 604 (e.g., a rotatable and depressible input mechanism such as, for example, input mechanism 506), and button 606 (e.g., a physical button such as, for example, input mechanism 508). In FIG. 6A, computer system 600 is a smartwatch. In some embodiments, computer system 600 is a smartphone, a laptop computer, a desktop computer, a tablet computer, or a head-mounted display device. In FIG. 6A, display 602 is a touch-sensitive display. In some embodiments, computer system 600 includes a touch-sensitive surface or other input device (e.g., a mouse and/or keyboard) that is separate from display 602.

In FIG. 6A, computer system 600 displays calculator user interface 608 of a calculator application on display 602. While displaying calculator interface 608, computer system 600 detects input 610a (e.g., a press of input mechanism 604, a voice command, or other input corresponding to a request to cease display of calculator user interface 608, close a currently-displayed application, exit a currently-displayed application, and/or move a currently-displayed application to a background process).

In some embodiments, any of the inputs described herein (e.g., input 610a, 610b, 610c, 610d, 610e, 610f, 610g, 610h, 610i, 610j, 610k, 610l, 610m, 610n, and/or 610o) is or includes a touch input (e.g., a tap gesture and/or a swipe gesture). In some embodiments, any of the inputs described herein (e.g., input 610a, 610b, 610c, 610d, 610e, 610f, 610g, 610h, 610i, 610j, 610k, 610l, 610m, 610n, and/or 610o) is or includes a voice input (e.g., a voice command to select a user interface element or to activate a feature or perform a function, such as a feature or function associated with a user interface element). In some embodiments, any of the inputs described herein (e.g., input 610a, 610b, 610c, 610d, 610e, 610f, 610g, 610h, 610i, 610j, 610k, 610l, 610m, 610n, and/or 610o) is or includes an air gesture (e.g., an air gesture to select a user interface element or to activate a feature or perform a function, such as a feature or function associated with a user interface element). In some embodiments, any of the inputs described herein (e.g., input 610a, 610b, 610c, 610d, 610e, 610f, 610g, 610h, 610i, 610j, 610k, 610l, 610m, 610n, and/or 610o) is or includes activation (e.g., a press, a rotation, and/or a movement) of a hardware device (e.g., a button, a rotatable input mechanism, a rotatable and depressible input mechanism, a mouse button, a button of a remote control, and/or a joystick). In some embodiments, any of the user interface elements described as being selected herein (e.g., an icon, affordance, button, and/or selectable option) is selected by activating a hardware device while the user interface element is in focus (e.g., highlighted, bolded, outlined, visually distinguished from other user interface elements, and/or located at or near a cursor).

In response to detecting input 610a, computer system 600 displays user interface 612 (e.g., a springboard user interface, a system user interface, or a home screen), as shown in FIG. 6B. User interface 612 includes selectable application icons for launching, visiting, opening, and/or displaying a user interface of respective applications. In FIG. 6B, computer system 600 detects selection 610b of stocks application object 612a, which corresponds to a stocks application (e.g., an application that displays stock market information and/or information related to financial assets such as stocks, bonds, mutual funds, commodities, and/or currencies).

In response to detecting selection 610b of stocks application object 612a, computer system 600 displays stocks application user interface 614 of a stocks application, as shown in FIG. 6C. While displaying stocks application user

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interface **614**, computer system detects a request (e.g., input **610c**) to display a set of application views (e.g., an application dock user interface and/or a user interface that displays the set of application views without displaying a user interface of an application). In FIG. 6C, input **610c** includes an input (e.g., a press and/or touch) on button **606**. In some embodiments, input **610c** includes a press of input mechanism **604**, a voice input, a touch on display **602**, or other input corresponding to a request to display a set of application views.

In response to detecting input **610c**, computer system **600** displays application view user interface **616** (e.g., an application dock), as shown in FIG. 6D. Application view user interface **616** displays a set of application views, which includes stocks application view **616a**, calculator application view **616b**, workout application view **616c**, and messages application view **616d** (which is described with reference to FIG. 6E). In FIG. 6D, the set of application views is arranged vertically relative to the orientation of display **602** in FIG. 6D. In some embodiments, the set of application views is arranged horizontally relative to the orientation of display **602** in FIG. 6D.

In some embodiments, an application view is a selectable user interface object that, when selected, displays a user interface of (e.g., opens and/or launches) a corresponding application. In some such embodiments, application view user interface **616** enables a user to quickly and easily access the applications corresponding to the displayed set of application views.

In some embodiments, computer system **600** displays the set of application views in an order that is based at least in part on the recency with which an application was displayed, used, viewed, visited, launched, and/or opened. In FIG. 6D, application views are displayed in descending order from top to bottom of display **602** as oriented in FIG. 6D, such that the application view that is first in the order (e.g., at the beginning of the order, at the top of the order, earliest in the order, and/or highest in the order) is displayed above application views that are lower in the order (e.g., the second application view in the order, the last application view in the order, and/or the application view at the end or bottom of the order).

As described herein, a position in the order of application views that is closer to the start of the order than another position is referred to as being “higher” in the order, “earlier” in the order, and/or “before” the other position in the order. For example, the first position in the order is before the second position in the order. Conversely, a position in the order of application views that is farther from the start of the order (e.g., closer to an end of the order) than another position is referred to as being “lower” in the order, “later” in the order, or “after” the other position in the order. For example, the second position in the order is after the first position in the order.

In FIG. 6D, since the stocks application corresponding to stocks application user interface **614** was the most recently visited application (e.g., the first most recently visited application), stocks application view **616a** is displayed in a first position (e.g., a top position) in application view user interface **616** (e.g., before or higher in the order of application views than calculator application view **616b**). Since the calculator application corresponding to calculator application user interface **608** was the next (e.g., the second) most recently visited application (e.g., prior to visiting the stocks application corresponding to stocks application user interface **614**), calculator application view **616b** is displayed in a second position (e.g., a middle position) in application

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view user interface **616** after (e.g., below or after) stocks application view **616a**. Since a workout application was the next (e.g., third) most recently visited application (e.g., prior to visiting the calculator application corresponding to calculator application user interface **608**), workout application view **616c** is displayed in a third position (e.g., a bottom position) in application view user interface **616** after (e.g., below) calculator application view **616b**. Notably, since user interface **612** is not an application user interface in the illustrated embodiment, the set of application views does not include a view corresponding to user interface **612** (e.g., computer system **600** does not display an application view corresponding to user interface **612**).

In some embodiments, computer system **600** only displays a subset (e.g., one, two, three, or four) (e.g., less than all) of the application views in the set of application views at a particular time. For example, in some embodiments, the set of application views includes one or more application views other than stocks application view **616a**, calculator application view **616b**, and workout application view **616c** (e.g., the set of application views includes four or more application views). In some embodiments, computer system **600** can display (e.g., reveal) other application views (e.g., hidden application views) in the set of application views in response to detecting an input. For example, in FIG. 6D, computer system **600** detects input **610d** (e.g., a swipe gesture, such as an upward swipe gesture, or other navigation input corresponding to a request to scroll the set of application views or otherwise display additional application views in the set of application views). Alternatively or in addition to input **610d**, computer system **600** detects input **610o**, which includes a rotation of input mechanism **604**.

In response to detecting input **610d** and/or **610o**, computer system **600** displays at least a portion of messages application view **616d** (e.g., by scrolling the set of application views). In some embodiments, in response to detecting rotation of input mechanism **604** in a first direction (e.g., a clockwise direction), computer system **600** scrolls the set of application views in a first direction (e.g., upward as oriented in FIG. 6D); and in response to detecting rotation of input mechanism **604** in a second direction (e.g., a counter-clockwise direction and/or an opposite direction), computer system **600** scrolls the set of application views in a second direction (e.g., downward as oriented in FIG. 6D and/or an opposite direction).

As shown in FIG. 6E, in response to detecting input **610d** and/or **610o**, computer system **600** ceases display of stocks application view **616a**, moves calculator application view **616b** from the second position on application view user interface **616** to the first position, moves workout application view **616c** from the third position on application view user interface **616** to the second position, and displays messages application view **616d** at the third position on application view user interface **616**. Although the application views have changed positions on application view user interface **616**, the order of the application views is maintained (e.g., does not change) in response to detecting input **610d** and/or **610o**. For example, in response to detecting a downward swipe and/or a rotation of input mechanism **604** in a direction that is opposite to the direction of input **610o** while displaying the set of application views as shown in FIG. 6E, computer system displays the set of application views as shown in FIG. 6D (e.g., by scrolling the set of application views downward).

In some embodiments, the workout application corresponding to workout application view **616c** can track the status of various types of workouts (e.g., walking, running,

swimming, and/or elliptical). In some embodiments, the workout application corresponding to workout application view **616c** can track the status of a workout (e.g., an active or ongoing workout) when a user interface of the workout application is not displayed (e.g., the workout application can run in the background and track an active workout). When the workout application corresponding to workout application view **616c** is actively tracking a workout, the workout application is referred to as being in an active session (e.g., regardless of whether or not a user interface of the workout application is currently displayed). In some embodiments, only certain applications are capable of being in an active session for the purpose of displaying application views. For example, although both the stocks application corresponding to stocks application view **616a** and the calculator application corresponding to calculator application view **616b** were visited, neither the stocks application nor the calculator application is performing an active function, and therefore neither the stocks application nor the calculator application is in an active session.

While computer system **600** is displaying application view user interface **616** as shown in FIG. 6E, the workout application corresponding to workout application view **616d** is not in an active session. Accordingly, in FIG. 6E, computer system **600** does not display a status indicator (e.g., status indicator **616c1** in FIG. 6J) that indicates a status of the workout application corresponding to workout application view **616d** (e.g., workout application view **616d** does not include a status indicator).

In some embodiments, computer system **600** can open an application in response to detecting selection of a corresponding application view. For example, in FIG. 6E, computer system **600** detects a selection (e.g., input **610e** and/or other selection input) of workout application view **616c**. In response to detecting the selection of workout application view **616c**, computer system **600** displays workout application user interface **618** of the workout application corresponding to workout application view **616c**, as shown in FIG. 6F. Workout application user interface **618** includes workout object **618a** corresponding to a first type of workout (e.g., an outdoor walk) and workout object **618b** corresponding to a second type of workout (e.g., an outdoor run). In FIG. 6F, computer system **600** detects selection (e.g., input **610f** and/or other selection input) of the first type of workout (e.g., selection of workout object **618a**). In response to detecting selection of the first type of workout, computer system **600** begins tracking a workout of the first type of workout and displays workout application user interface **620**, which displays information associated with the workout, as shown in FIG. 6G. Accordingly, in FIG. 6G, the workout application is in an active session.

While the workout application corresponding to workout application user interface **620** is in an active session, computer system **600** detects input **610g** (e.g., a press of input mechanism **604** and/or other selection input). In response to detecting input **610g**, computer system **600** displays user interface **612**, as shown in FIG. 6H. User interface **612** is described above with reference to FIG. 6B. In the embodiment illustrated in FIG. 6H, the workout application corresponding to workout application user interface **620** remains in an active session (e.g., continues tracking the workout) while computer system **600** displays user interface **612** (e.g., while computer system **600** is not displaying a user interface of the workout application, such as workout application user interface **618** or workout application user interface **620**).

While displaying user interface **612** in FIG. 6H, computer system **600** detects a request (e.g., input **610h** and/or other

selection input) to open a photos application corresponding to photos application object **612b**. In response to detecting the request to open the photos application, computer system **600** displays photos application user interface **622** corresponding to photos application object **612b**, as shown in FIG. 6I. While displaying photos application user interface **622**, computer system **600** detects a request (e.g., input **610i** and/or other selection input) to display a set of application views. In response to detecting the request to display a set of application views, computer system **600** displays (e.g., re-displays or re-opens) application view user interface **616**, as shown in FIG. 6J. Although the request to display a set of application views in FIG. 6I is detected while displaying a user interface of an application, in some embodiments, a request to display a set of application views can be detected while displaying other types of user interfaces (e.g., a home screen, a system user interface, or a springboard user interface). For example, in some embodiments, computer system **600** displays application view user interface **616** in response to detecting a request (e.g., a press of input mechanism **606** and/or other input) to display a set of application views while displaying user interface **612**.

In some embodiments, application view user interface **616** displays a set of application views based on application activity status and application usage. In some embodiments, computer system **600** displays applications views of applications that are in an active session before (e.g., earlier in the order of application views than, higher in the order of application views that, above, and/or to the left of) application views corresponding to the most recently visited applications that are not in an active session. For example, in FIG. 6J, because the workout application corresponding to workout application view **616c** is in an active session, workout application view **616c** is first in (e.g., at the beginning of) the order of application views and is displayed in the first position in application view user interface **616**. Since the photos application was the most recently visited application that is not in an active session, photos application view **616e** is second in the order of application views (e.g., after workout application view **616c**), and is displayed in the second position on application view user interface **616** (e.g., even though the photos application corresponding to photos application view **616c** was visited more recently than the workout application corresponding to workout application view **616c**). Since the stocks application was the next most recently visited application that is not in an active session prior to the photos application, stocks application view **616a** is third in the order of application views (e.g., after photos application view **616e**), and is displayed in the third position on application view user interface **616**.

In FIG. 6J, since the workout application corresponding to workout application view **616d** is in an active session, computer system **600** displays status indicator **616c1** (e.g., active), which indicates a status (e.g., a current status and/or a session status) of the workout application corresponding to workout application view **616d**. In FIG. 6J, status indicator **616c1** is displayed on (e.g., included in) workout application view **616c**.

While displaying application view user interface **616** in FIG. 6J, computer system **600** detects input **610j** (e.g., a press of button **606** or other navigation or selection input). In some embodiments, input **610j** corresponds to a request to cease display of the set of application views and/or application view user interface **616** (e.g., to exit application view user interface **616**). In some embodiments, input **610j** is a request to return to the user interface that was displayed immediately prior to displaying application view user inter-

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face **616** in FIG. 6J (e.g., the user interface that was displayed at the time of receiving the request to display application view user interface **616** in FIG. 6J). In response to detecting input **610j**, computer system **600** displays (e.g., returns to) photos application user interface **622**, as shown in FIG. 6K. In the embodiment illustrated in FIG. 6K, the workout application corresponding to workout application user interface **620** remains in an active session (e.g., continues tracking the workout) while computer system **600** displays photos application user interface **622** (e.g., while visiting the photos application and/or while the photos application is open).

In FIG. 6K, while displaying photos application user interface **622**, computer system **600** detects a request (e.g., input **610k** or other navigation or selection input) to cease display of photos application user interface **622** (e.g., to exit and/or close the photos application). In FIG. 6K, input **610k** includes a press of input mechanism **604**.

In response to detecting the request to cease display of photos application user interface **622**, computer system **600** displays user interface **612**, as shown in FIG. 6L. User interface **612** is described above with reference to FIG. 6B. In the embodiment illustrated in FIG. 6L, the workout application corresponding to workout application user interface **620** remains in an active session (e.g., continues tracking the workout) while computer system **600** displays user interface **612** (e.g., while computer system **600** is not displaying a user interface of the workout application, such as workout application user interface **618** or workout application user interface **620**).

While displaying user interface **612** in FIG. 6L, computer system **600** detects a request (e.g., input **610l** and/or other selection input) to open a media application (e.g., a music application, video application, and/or multimedia application) corresponding to media application object **612c**. In response to detecting the request to open the media application, computer system **600** displays media application user interface **624** corresponding to media application object **612c**, as shown in FIG. 6M.

In FIG. 6M, the media application corresponding to media application user interface **624** is not in an active session (e.g., is not playing, recording, or navigating media such as a song, podcast, video, or voice recording). The media application corresponding to media application user interface **624** has also not been recently visited (e.g., as determined by recency criteria). Accordingly, an application view corresponding to the media application is not displayed or included in the sets of application views described with reference to FIGS. 6D-6E and 6J.

While displaying media application user interface **624** in FIG. 6M, computer system **600** detects a request (e.g., input **610m** and/or other selection input on playback control object **624a**) to begin playback of media content. In response to detecting the request to begin playback of media content, computer system **600** begins playback of media content, as indicated by the change in playback control object **624a** from FIG. 6M to FIG. 6N. In FIG. 6N, since the media application is actively playing media content, the media application is in an active session (e.g., an active media playback session).

While displaying media application user interface **624** in FIG. 6N, computer system **600** detects a request (e.g., input **610n** on input mechanism **606** and/or other selection input) to display a set of application views (e.g., to display application view user interface **616**). In response to detecting the request to display a set of application views, computer system **600** displays (e.g., re-opens or re-displays) applica-

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tion view user interface **616** as shown in FIG. 6O. In FIG. 6O, application view user interface **616** displays a set of application views, including media application view **616f**, workout application view **616c**, and photos application view **616e**.

In some embodiments, computer system **600** displays applications views of applications that are in an active session in an order according to how recently the applications that are in an active session were visited, followed by application views of applications (e.g., recently visited applications and/or designated applications) that are not in an active session in an order according to how recently the applications that are not in an active session were visited. In some embodiments, computer system **600** displays applications views of applications that are in an active session (e.g., applications views for all applications that are in an active session) earlier in the order of application views than application views (e.g., any application view) of applications (e.g., recently visited applications and/or designated applications) that are not in an active session. For example, computer system **600** can group application views of applications (e.g., all applications) that are in an active session at the beginning of the order of application views, followed by application views of applications that are not in an active session (e.g., and that meet recency and/or designated criteria). In some embodiments, computer system **600** does not (or will not) display an application view of an application that is in an active session in a position in the order of application views that is after a position of an application view of an application that is not in an active session. In other words, in some embodiments, computer system **600** does not (or will not) display an application view of an application that is not in an active session in a position in the order of application views that is before a position of an application view of an application that is in an active session.

For example, in FIG. 6O, because the media application corresponding to media application view **616f** is in an active session and was visited more recently than the workout application corresponding to workout application view **616c** (which is also in an active session), computer system **600** places media application view **616f** before workout application view **616c** in the order of application views (e.g., media application view **616f** is first, or at the beginning, in the order of application views) and displays media application view **616f** in the first position in application view user interface **616**.

Because the workout application corresponding to workout application view **616c** is still in an active session, workout application view **616c** is second in the order of application views (e.g., before any application views of applications that are not in an active session) and is displayed in the second position in application view user interface **616**. Because there are no other applications in an active session besides the applications corresponding to media application view **616f** and workout application view **616c**, and because the photos application corresponding to photos application view **616e** is the most recently visited application that is not in an active session, computer system **600** places photos application view **616e** third in the order of application views and displays photos application view **616e** in the third position in application view user interface **616**.

In FIG. 6O, since the media application corresponding to media application view **616f** is in an active session, computer system **600** displays status indicator **616/1** (e.g., playing), which indicates a status (e.g., a current status and/or an active session status) of the media application corresponding

to media application view **616f**. In FIG. **6O**, status indicator **616i** is displayed on (e.g., included in) media application view **616f**. Since the workout application corresponding to workout application view **616c** is in (e.g., still in) an active session, computer system **600** displays status indicator **616c1** (e.g., active), which indicates a status (e.g., a current status and/or an active session status) of the workout application corresponding to workout application view **616c**.

FIG. **7** is a flow diagram illustrating a method for displaying a set of application views using a computer system in accordance with some embodiments. Method **700** is performed at a computer system (e.g., **100**, **300**, **500**, **600**, a smartphone, a smartwatch, a tablet computer, a desktop computer, a laptop computer, and/or a head-mounted device (e.g., a head-mounted augmented reality and/or extended reality device)) that is in communication with (e.g., includes and/or is connected to) a display generation component (e.g., **112**, **602**, a monitor, a display, a display controller, a touch-sensitive display system, a touchscreen, a touch-sensitive display, a projector, a holographic display, and/or a head-mounted display system) and one or more input devices (e.g., **602**, **604**, **606**, a touch-sensitive surface, a keyboard, a mouse, a joystick, a trackpad, a microphone, one or more optical sensors for detecting gestures, one or more capacitive sensors for detecting hover inputs, an accelerometer, a gyroscope, and/or an inertial measurement unit). Some operations in method **700** are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted.

As described below, method **700** provides an intuitive way for displaying a set of application views. The method reduces the cognitive burden on a user for displaying a set of application views, thereby creating a more efficient human-machine interface. For battery-operated computing devices, enabling a user to display a set of application views and access applications faster and more efficiently conserves power and increases the time between battery charges.

The computer system receives (**702**) (e.g., detects), via the one or more input devices, a request (e.g., **610c**, **610i**, **610n**, and/or an input corresponding to a request) to display a set of application views (e.g., **616a**, **616b**, **616c**, **616d**, **616e**, and/or **616f**). In some embodiments, an application view includes (or is) a representation of an application, a graphical object corresponding to an application (e.g., a selectable user-interactive graphical user interface object for opening or launching an application), an icon corresponding to an application (e.g., a selectable icon for opening or launching an application), a screenshot of a user interface of an application, an image corresponding to an application and/or of a user interface of an application, and/or a thumbnail image corresponding to an application and/or of a user interface of an application. In some embodiments, an application view displays a most-recently displayed view of a user interface of an application. In some embodiments, an application view displays a snapshot or screenshot of a user interface of an application as it was most recently displayed, such as at the time the application was closed, at the time of switching from the application to another application and/or user interface (e.g., a home screen or watch face), and/or at the time of receiving the request to display the set of application views. In some embodiments, an application view displays a current state, status, and/or information of an application and/or a user interface of an application. For example, an application view can display a status and/or information of a corresponding workout application that is running (e.g., in the background) and can update over time to display current or more recent status and/or information.

In response to receiving the request to display a set of application views, the computer system displays (**704**), via the display generation component, a set of (e.g., one or more) application views (e.g., **616**, application views of a subset of the applications installed on the computer system), the set of application views including a first application view (e.g., **616a**, **616b**, **616c** in FIGS. **6D** and **6E**, **616d**, or **616e**) that corresponds to a first application (e.g., a preselected application, a designated application (such as an application designated as a favorite application), a most recently used application, an application that was most recently opened, launched, viewed, and/or visited on, by, or using the computer system, and/or an application that is within a threshold number of applications most recently opened, launched, viewed, and/or visited on, by, or using the computer system) that is not in an active session. In some embodiments, an application is in an active session when the application is actively performing a predefined function or predefined operation such as playing media content (e.g., music, an audiobook, a video, a movie, a voice recording, and/or a podcast), monitoring a status of an activity (e.g., a workout, a pace, a distance, repetitions, steps, strides, and/or a mindfulness session), measuring a biometric characteristic (e.g., blood pressure, heart rate, pulse, blood oxygen level, and/or electrocardiogram), performing a navigation function, remotely controlling an external device, providing a live communication session (e.g., a phone call, a video chat, or a walkie talkie communication), and/or counting or time-keeping (e.g., running a stopwatch or a timer). In some embodiments, an application (e.g., a calculator application, a stocks application, or a photos application) is not in an active session when the application is open (e.g., running in the background) but is not actively performing a predefined function (e.g., a function other than a process that is required to run the application (or keep the application running) in the background). In some embodiments, a navigation application is in an active session when there is an active route (e.g., monitoring a location of a device or user to a destination). In some embodiments, a navigation application is in an active session when the application is actively monitoring and/or updating location information of a user and/or device. In some embodiments, a live communication application (e.g., real-time communication application, a phone application, a video chat application, and/or an audio/video communication application) is in an active session during an active or ongoing communication session (e.g., a live communication session, a live audio communication session, a live video communication session, a live audio/video communication session, an active phone call, an active video chat session). In some embodiments, a live communication application is not in an active session when there is not an active or ongoing communication session. In some embodiments, an audio communication application is in an active session when providing a walkie talkie function, recording audio (e.g., recording a voice memo), and/or playing recorded audio (e.g., playing a voice memo). In some embodiments, an audio communication application is in an active session when providing a walkie talkie function, recording audio (e.g., recording a voice memo), and/or playing recorded audio (e.g., playing a voice memo). In some embodiments, an application is in an active session when the application is actively controlling an external device and/or being used to remotely control an external device that is actively performing an operation. The set of application views are arranged in an order (e.g., a sequence) of application views.

Displaying the set of application views includes: in accordance with a determination that a second application is in an

active session (e.g., on the computer system), displaying (706) a second application view (e.g., 616c in FIGS. 6J and 6O, or 616f in FIG. 6O) that corresponds to the second application before (e.g., earlier than, higher than, closer to the beginning of the order of application views than, ahead of, in front of, and/or above) the first application view in the order of application views (e.g., 616c is displayed before 616a and 616e in FIG. 6J; 616c is displayed before 616e in FIG. 6O; 616f is displayed before 616e in FIG. 6O); and in accordance with a determination that there are no applications in an active session (e.g., on the computer system; as shown in FIG. 6C), displaying (708) the first application view in a first position (e.g., a top position, a starting position, a beginning position, an initial position, and/or a left-most position) in the order of application views (e.g., 616a is displayed in the first position in the order of application views in FIG. 6D). In some embodiments, the application corresponding to the first application view was opened, launched, viewed, and/or visited more recently than the application that is in the active session (e.g., the second application view that corresponds to the application that is in the active session is displayed before the first application view in the order of application views even if the application corresponding to the first application view was opened, launched, viewed, and/or visited more recently than the application that is in the active session).

Displaying the second application view of the second application before the first application view in the order of application views in accordance with a determination that the second application is in an active session prioritizes application views of applications that are in an active session, which are likely to be particularly relevant to a user at the time of the request to display the set of application views and makes applications that are in an active session easier to access. Prioritizing application views of applications that are in an active session provides improved visual feedback to the user (e.g., by displaying more relevant application views), reduces the number of inputs needed to perform an operation (e.g., to access applications that are in an active session), and performs an operation when a set of conditions has been met without requiring further user input.

In some embodiments, displaying the second application view that corresponds to the second application includes displaying a status indicator (e.g., 616c1 and/or 616f1) (e.g., concurrently with and/or on an application view that corresponds to the second application) that indicates a status (e.g., active, playing, recording, and/or connected) of the second application (and/or a status of the active session). Displaying a status indicator that indicates the status of the second application indicates to the user that the second application is in an active session and provides the user with additional information about the state of the second application and/or the active session, which provides improved visual feedback to the user.

In some embodiments, while displaying the set of application views: in accordance with a determination that a respective application is in an active session, the computer system displays a status indicator (e.g., 616c1 and/or 616f1) (e.g., concurrently with and/or on an application view (e.g., 616c in FIG. 6J or 6O and/or 616f in FIG. 6O) that corresponds to the respective application) that indicates a status (e.g., active, playing, recording, and/or connected) of the respective application (and/or the status of the active session); and in accordance with a determination that the respective application is not in an active session (and, optionally, in accordance with a determination that the respective application satisfies recency criteria or designated-application criteria), the computer system displays an application view (e.g., 616c in FIG. 6D or FIG. 6E) that corresponds to the respective application without displaying the status indicator. In some embodiments, recency criteria includes a criterion that is satisfied if an application is one of the N-most recently used or visited applications, where N is, e.g., one, two, three, four, five, six, seven, eight, nine, or ten (e.g., the computer system displays an application view of the respective application in the set of application views if (or only if) the respective application is one of the N-most recently used applications). In some embodiments, a designated-application criteria includes a criterion that is satisfied if an application (or an application view) has been (or currently is) designated to be included in the set of application views (e.g., a user has designated the application or application view as a favorite). Displaying a status indicator for a respective application based on whether or not the respective application is in an active session indicates to the user whether or not the respective application is in an active session indicates to the user that the second application is in an active session, and provides the user with additional information about the state of the respective application and/or the active session, without requiring the user to provide further input (e.g., to open the respective application to determine the state of the application), provides improved visual feedback to the user, reduces the number of inputs needed to perform an operation, and performs an operation when a set of conditions has been met without requiring further user input.

In some embodiments, displaying the set of application views includes displaying a third application view (e.g., 616a, 616b, 616c, 616d, 616e, 616f, the first application view or the second application view) that corresponds to a third application (e.g., the first application or the second application). In some embodiments, displaying the third application view includes: in accordance with a determination that the third application is in an active session, displaying the third application view in a first respective position in the order of application views (e.g., 616c in FIG. 6J or 6O); and in accordance with a determination that the third application is not in an active session (and, optionally, in accordance with a determination that the third application satisfies recency criteria or designated-application criteria), displaying the third application view in a second respective position in the order of application views (e.g., 616c in FIGS. 6D and 6E), wherein the second respective position in the order of application view is after the first respective position in the order of application views. In some embodiments, the first respective position is before all application views in the set of application views that correspond to applications that are not in an active session (e.g., there are no application views of applications that are not in an active session that are displayed in a position that is before the first respective position). In some embodiments, recency criteria includes a criterion that is satisfied if an application is one of the N-most recently used or visited applications, where N is, e.g., one, two, three, four, five, six, seven, eight, nine, or ten (e.g., the computer system displays an application view of the respective application in the set of application views if (or only if) the respective application is one of N-most recently used applications). In some embodiments, a designated-application criteria includes a criterion that is satisfied if an application (or an application view) has been (or currently is) designated to be included in the set of application views (e.g., a user has designated the application or application view as a favorite). Displaying the third application view of the third application in a higher position (e.g.,

the computer system displays an application view (e.g., 616c in FIG. 6D or FIG. 6E) that corresponds to the respective application without displaying the status indicator. In some embodiments, recency criteria includes a criterion that is satisfied if an application is one of the N-most recently used or visited applications, where N is, e.g., one, two, three, four, five, six, seven, eight, nine, or ten (e.g., the computer system displays an application view of the respective application in the set of application views if (or only if) the respective application is one of the N-most recently used applications). In some embodiments, a designated-application criteria includes a criterion that is satisfied if an application (or an application view) has been (or currently is) designated to be included in the set of application views (e.g., a user has designated the application or application view as a favorite). Displaying a status indicator for a respective application based on whether or not the respective application is in an active session indicates to the user whether or not the respective application is in an active session indicates to the user that the second application is in an active session, and provides the user with additional information about the state of the respective application and/or the active session, without requiring the user to provide further input (e.g., to open the respective application to determine the state of the application), provides improved visual feedback to the user, reduces the number of inputs needed to perform an operation, and performs an operation when a set of conditions has been met without requiring further user input.

closer to the start) in the order of application views when the third application is active than when the third application is not active allows the third application view to be displayed regardless of whether or not the third application is in an active session, and prioritizes the third application view when the third application is in an active session, which provides improved visual feedback to the user (e.g., by displaying more placing the third application view in a more prominent position when the third application is in an active session), reduces the number of inputs needed to perform an operation (e.g., to access the third application when it is in an active session), and performs an operation when a set of conditions has been met without requiring further user input.

In some embodiments, while displaying the second application view (e.g., **616c** or **616f**) that corresponds to the second application: the computer system displays a fourth application view (e.g., **616c** or **616f**) that corresponds to a fourth application, wherein the fourth application is in an active session. In some embodiments, displaying the fourth application view includes displaying the fourth application view before the first application view (e.g., **616e** in FIG. 6O) in the order of application views (e.g., in accordance with a determination that the fourth application is in an active session), including: in accordance with a determination that the fourth application was visited (e.g., used, opened, launched, and/or accessed) more recently than the second application, displaying the fourth application view before the second application view in the order of application views (e.g., **616f** is displayed before **616c** in FIG. 6O since the media application was used more recently than the workout application); and in accordance with a determination that the second application was visited (e.g., used, opened, launched, and/or accessed) more recently than the fourth application, displaying the second application view before the fourth application view in the order of application views (e.g., if the workout application was used more recently than the media application, then **616c** is displayed before **616f** in FIG. 6O). Displaying the fourth application view and the second application view in a relative order based on which corresponding application was used more recently enables the computer system to prioritize the display of application views of applications that are in an active session based on recency of use, which provides improved visual feedback to the user (e.g., by displaying the application view of the more recently used application in a higher position in the order of application views), reduces the number of inputs needed to perform an operation (e.g., to access the more recently used application), and performs an operation when a set of conditions has been met without requiring further user input.

In some embodiments, displaying the set of application views includes displaying the first application view without displaying a fifth application view that corresponds to a fifth application (e.g., and that is in the set of application views) (e.g., displaying **616a**, **616b**, and/or **616c** without displaying **616d**, as shown in FIG. 6D). In some embodiments, while displaying the first application view without displaying the fifth application view, the computer system detects, via the one or more input devices, a first input (e.g., **610d**, **610o**, a swipe gesture, a rotation of a rotatable input mechanism, and/or a request to scroll the set of application views). In some embodiments, in response to detecting the first input, the computer system displays the fifth application view (e.g., by scrolling the set of application views) (e.g., display **616d** in FIG. 6E). Displaying the fifth application view in response to detecting input provides the user with access to additional application views and allows the computer system to display application views with a larger size (e.g., since

additional application views can accessed through user input instead of having to display the entire set of application views at one time), which provides improved visual feedback to the user.

In some embodiments, displaying the set of application views includes concurrently displaying the first application view and a sixth application view (e.g., the second application view) that at least partially overlaps the first application view (e.g., the first application view is displayed in front of (e.g., blocks, covers, or obstructs) at least a portion of the sixth application view, or the sixth application view is displayed in front of at least a portion the first application view; **616a** and **616b** overlap in FIG. 6D; **616b** and **616c** overlap in FIGS. 6D and 6E; **616c** and **616d** overlap in FIG. 6E; **616c** and **616e** overlap in FIG. 6J; **616e** and **616a** overlap in FIG. 6J; **616f** and **616c** overlap in FIG. 6O; and **616c** and **616e** overlap in FIG. 6O). Concurrently displaying the first application view and a sixth application view that at least partially overlaps the first application view enables the computer system to display more application views at a time and reduces the number of inputs required to access application views (e.g., by reducing the need to for scrolling and/or navigating to display additional application views), which provides improved visual feedback to the user, reduces the number of inputs needed to perform an operation, and provides additional control options without cluttering the user interface with additional displayed controls.

In some embodiments, while displaying the first application view (e.g., **616a**, **616b**, **616c**, **616d**, **616e**, and/or **616f**), the computer system detects an input (e.g., **610e**) corresponding to a selection of the first application view (e.g., a tap on the first application view or an activation of a button while the first application view is designated or in focus). In some embodiments, in response to detecting the input corresponding to selection of the first application view, the computer system displays (e.g., initiates display of) a user interface (e.g., **618**) of the first application (e.g., launches the first application, opens the first application, and/or displays the user interface of the first application in a foreground). In some embodiments, displaying the user interface of the first application includes displaying the user interface of the first application in a most-recent previously displayed state. In some embodiments, while concurrently displaying the first application view and a seventh application view (e.g., any application view in the set of application views other than the first application view), the computer system detects an input, and in response to detecting the input: in accordance with a determination that the input corresponds to selection of the first application view, the computer system displays a user interface of the first application (e.g., without displaying a user interface of the seventh application); and in accordance with a determination that the input corresponds to selection of the seventh application view, the computer system displays a user interface of the seventh application (e.g., without displaying a user interface of the first application). Displaying a user interface of the first application in response to detecting input corresponding to selection of the first application view allows a user to quickly and easily access an application by, e.g., requesting display of the set of application views and then selecting an application view corresponding to a desired application, which provides improved visual feedback to the user and reduces the number of inputs needed to perform an operation (e.g., to access an application).

In some embodiments, the first application view includes a representation (e.g., an image, a snapshot, a screenshot, and/or a thumbnail image) of a user interface of the first

application (e.g., **616a**, **616b**, and/or **616c** in FIG. 6D; **616b**, **616c**, and/or **616d** in FIG. 6E; **616c**, **616e**, and/or **616a** in FIG. 6J; and **616f**, **616c**, and/or **616e** in FIG. 6O). In some embodiments, an application view includes a representation of a most-recently displayed state of the user interface of the respective application. In some embodiments, the most-recently displayed state of an application is the last state of the application before closing the application and/or changing from the user interface of the respective application to a user interface that does not correspond to the respective application, such as, e.g., a user interface of a different application, a user interface that includes the set of application views, and/or a system user interface (e.g., a home screen, a springboard user interface, or a clock user interface). Including a representation of a user interface of a respective application in the application view that corresponds to the respective application provides the user with additional information about the respective application, such as a state of the application, and enables the user to more easily and accurately identify the application that corresponds to a respective application view, which provides improved visual feedback to the user and reduces the number of inputs needed to perform an operation (e.g., by reducing the number of erroneous inputs).

In some embodiments, the request to display the set of application views includes (e.g., is) a press (e.g., **610a**, **610c**, **610g**, **610i**, **610j**, and/or **610n**) of a physical button (e.g., **604**, **606**, a physical button on the computer system, such as, e.g., a button on a side of the computer system that is perpendicular to a surface of the display generation component). In some embodiments, the request to display the set of application views is detected while displaying a user interface of an application or a system user interface (e.g., a home screen, a springboard user interface, or a clock face user interface). For example, the computer system can initiate display of the set of application views from an application user interface, a home screen, an application springboard, a system settings user interface, or a clock face user interface (e.g., on a smartwatch). Displaying the set of application views in response to a request that includes a press of a physical button allows the user to access the set of application views without requiring additional inputs to locate and/or navigate to a particular user interface or user interface object (e.g., the user can directly access the set of application views regardless of the user interface that is currently displayed), which provides improved visual feedback to the user and reduces the number of inputs needed to perform an operation.

In some embodiments, the request (e.g., **610c**, **610i**, or **610n**) to display the set of application views is received while displaying a first respective user interface (e.g., **608**, **612**, **614**, **618**, **620**, **622**, **624**, an application user interface, a home screen, an application springboard, a system settings user interface, or a clock face user interface (e.g., on a smartwatch)). In some embodiments, while displaying the set of application views, the computer system detects, via the one or more input devices, a second input (e.g., **610a**, **610c**, **610g**, **610i**, **610j**, **610k**, or **610n**). In some embodiments, in response to detecting the second input, the computer system displays the first respective user interface (e.g., **608**, **612**, **614**, **618**, **620**, **622**, or **624**) (and, optionally, ceasing display of the set of application views). In some embodiments, displaying the first respective user interface includes launching or opening an application corresponding to the first respective user interface. Displaying a user interface that was displayed at the time of receiving the request to display the set of application views in response

detecting an input while displaying the set of application views allows a user to quickly and easily return to a previously displayed user interface and/or switch between display of a user interface and display of the set of application views, which provides improved visual feedback to the user and reduces the number of inputs needed to perform an operation.

In some embodiments, displaying the set of application views includes concurrently displaying in a second respective user interface (e.g., **616** and/or an application dock user interface): a seventh application view (e.g., **616a**, **616b**, **616c**, **616d**, **616e**, **616f**, the first application view, or the second application view) in the first position (e.g., the starting position, the beginning position, the position corresponding to the displayed position of **616a** in FIG. 6D, the position corresponding to the displayed position of **616c** in FIG. 6J, or the position corresponding to the displayed position of **616f** in FIG. 6O) in the order of application views; and an eighth application view (e.g., **616a**, **616b**, **616c**, **616d**, **616e**, **616f**, the first application view, or the second application view) in the second position (e.g., immediately after the first position, the position corresponding to the displayed position of **616b** in FIG. 6D, the position corresponding to the displayed position of **616e** in FIG. 6J, or the position corresponding to the displayed position of **616c** in FIG. 6O) in the order of application views. In some embodiments, the eighth application view is displayed at a location in a center of the second respective user interface (e.g., the eighth application view is displayed in a center of the second respective user interface and/or is centered on the second respective user interface; the seventh application view is displayed towards a top of the second respective user interface relative to the eighth application view). In some embodiments, the seventh application view is displayed closer to an edge (e.g., a top edge) of the second respective user interface than the eighth application view. In some embodiments, the eighth application view overlaps (e.g., blocks, is displayed in front of, and/or obscures) at least a portion of the seventh application view. In some embodiments, in accordance with a determination that seventh application view corresponds to an application that is in an active session (and, optionally, that the request to display the set of application views is received while displaying a user interface that does not correspond to the seventh application view), the computer system displays the seventh application view (e.g., instead of the eighth application view) at the location in the center of the second respective user interface. Displaying the eighth application view (which is in the second position in the order of application views) at a location in the center of a user interface allows the eighth application view to be displayed in a prominent location on the user interface and enables the user to quickly and easily access the application that corresponds to the eighth application view and switch back and forth between the application that corresponds to the eighth application view and the application that corresponds to the seventh application view, which provides improved visual feedback to the user and reduces the number of inputs needed to perform an operation.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in

the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources to provide users with more relevant applications and user interfaces. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, social network IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to provide more relevant applications and user interfaces. Accordingly, use of such personal information data enables users to have access to more relevant applications and user interfaces. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should

be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of displaying application views, the present technology can be configured to allow users to select to "opt in" or "opt out" of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide data for displaying application views. In yet another example, users can select to limit the length of time data is maintained or entirely prohibit the development of a baseline profile. In addition to providing "opt in" and "opt out" options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, application views can be displayed by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the computer system, or publicly available information.

What is claimed is:

1. A computer system configured to communicate with a display generation component and one or more input devices, comprising:

one or more processors; and

memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for:

receiving, via the one or more input devices, a set of one or more inputs corresponding to a request to display a user interface of a respective application; in response to receiving the set of one or more inputs, displaying the user interface of the respective application;

while displaying the user interface of the respective application, receiving, via the one or more input devices, a request to display a set of application views; and

in response to receiving the request to display the set of application views, displaying, via the display generation component, the set of application views, wherein the set of application views includes a plurality of application views that includes a first application view that corresponds to a first application that is not in an active session and a second application view that corresponds to a second application, wherein:

- the first application and the second application are different from the respective application;
- an application that is actively performing one or more predefined functions is in an active session;
- an application that is not actively performing one or more of the one or more predefined functions is not in an active session;
- the set of application views are arranged in an order of application views based on application activity status and application usage; and
- displaying the set of application views includes:
 - in accordance with a determination that the second application is in an active session:
 - displaying the second application view that corresponds to the second application in a position before the first application view in the order of application views; and
 - displaying the first application view in a position that is subsequent to the second application view in the order of application views; and
 - in accordance with a determination that there are no applications in an active session, wherein the determination that there are no applications in an active session includes a determination that the first application is not in an active session:
 - displaying the first application view in a position before the second application view in the order of application views; and
 - displaying the second application view in a position that is subsequent to the first application view in the order of application views.

2. The computer system of claim 1, wherein displaying the second application view that corresponds to the second application includes displaying a status indicator that indicates a status of the second application.

3. The computer system of claim 1, wherein the one or more programs further include instructions for:

- while displaying the set of application views:
 - in accordance with a determination that a particular application is in an active session, displaying a status indicator that indicates a status of the particular application; and
 - in accordance with a determination that the particular application is not in an active session, displaying an application view that corresponds to the particular application without displaying the status indicator.

4. The computer system of claim 1, wherein displaying the set of application views includes displaying a third application view that corresponds to a third application, wherein displaying the third application view includes:

- in accordance with a determination that the third application is in an active session, displaying the third application view in a first respective position in the order of application views; and

in accordance with a determination that the third application is not in an active session, displaying the third application view in a second respective position in the order of application views, wherein the second respective position in the order of application view is after the first respective position in the order of application views.

5. The computer system of claim 1, wherein the one or more programs further include instructions for:

- while the second application view that corresponds to the second application is displayed:
 - displaying a fourth application view that corresponds to a fourth application, wherein the fourth application is in an active session, and wherein displaying the fourth application view includes displaying the fourth application view before the first application view in the order of application views, including:
 - in accordance with a determination that the fourth application was visited more recently than the second application, displaying the fourth application view before the second application view in the order of application views; and
 - in accordance with a determination that the second application was visited more recently than the fourth application, displaying the second application view before the fourth application view in the order of application views.

6. The computer system of claim 1, wherein displaying the set of application views includes displaying the first application view without displaying a fifth application view that corresponds to a fifth application and wherein the one or more programs further include instructions for:

- while displaying the first application view without displaying the fifth application view, detecting, via the one or more input devices, a first input; and
- in response to detecting the first input, displaying the fifth application view.

7. The computer system of claim 1, wherein displaying the set of application views includes concurrently displaying the first application view and a sixth application view that at least partially overlaps the first application view.

8. The computer system of claim 1, wherein the one or more programs further include instructions for:

- while displaying the first application view, detecting an input corresponding to a selection of the first application view; and
- in response to detecting the input corresponding to selection of the first application view, displaying a user interface of the first application.

9. The computer system of claim 1, wherein the first application view includes a representation of a user interface of the first application.

10. The computer system of claim 1, wherein the request to display the set of application views includes a press of a physical button.

11. The computer system of claim 1, wherein the request to display the set of application views is received while displaying a first respective user interface and wherein the one or more programs further include instructions for:

- while displaying the set of application views, detecting, via the one or more input devices, a second input; and
- in response to detecting the second input, displaying the first respective user interface.

12. The computer system of claim 1, wherein displaying the set of application views includes concurrently displaying in a second respective user interface:

a seventh application view in a first respective position in the order of application views; and
 an eighth application view in a second respective position in the order of application views, wherein the eighth application view is displayed at a location in a center of the second respective user interface.

13. A non-transitory computer-readable storage medium storing one or more programs configured to be executed by one or more processors of a computer system that is in communication with a display generation component and one or more input devices, the one or more programs including instructions for:

receiving, via the one or more input devices, a set of one or more inputs corresponding to a request to display a user interface of a respective application;

in response to receiving the set of one or more inputs, displaying the user interface of the respective application;

while displaying the user interface of the respective application, receiving, via the one or more input devices, a request to display a set of application views; and

in response to receiving the request to display the set of application views, displaying, via the display generation component, the set of application views, wherein the set of application views includes a plurality of application views that includes a first application view that corresponds to a first application that is not in an active session and a second application view that corresponds to a second application, wherein:

the first application and the second application are different from the respective application;

an application that is actively performing one or more predefined functions is in an active session;

an application that is not actively performing one or more of the one or more predefined functions is not in an active session;

the set of application views are arranged in an order of application views based on application activity status and application usage; and

displaying the set of application views includes:

in accordance with a determination that a second application is in an active session:

displaying the second application view that corresponds to the second application in a first position before the first application view in the order of application views; and

displaying the first application view in a position that is subsequent to the second application view in the order of application views; and

in accordance with a determination that there are no applications in an active session, wherein the determination that there are no applications in an active session includes a determination that the first application is not in an active session:

displaying the first application view in a position before the second application view in the order of application views; and

displaying the second application view in a position that is subsequent to the first application view in the order of application views.

14. The non-transitory computer-readable storage medium of claim 13, wherein displaying the second application view that corresponds to the second application includes displaying a status indicator that indicates a status of the second application.

15. The non-transitory computer-readable storage medium of claim 13, wherein the one or more programs further include instructions for:

while displaying the set of application views:

in accordance with a determination that a particular application is in an active session, displaying a status indicator that indicates a status of the particular application; and

in accordance with a determination that the particular application is not in an active session, displaying an application view that corresponds to the particular application without displaying the status indicator.

16. The non-transitory computer-readable storage medium of claim 13, wherein displaying the set of application views includes displaying a third application view that corresponds to a third application, wherein displaying the third application view includes:

in accordance with a determination that the third application is in an active session, displaying the third application view in a first respective position in the order of application views; and

in accordance with a determination that the third application is not in an active session, displaying the third application view in a second respective position in the order of application views, wherein the second respective position in the order of application view is after the first respective position in the order of application views.

17. The non-transitory computer-readable storage medium of claim 13, wherein the one or more programs further include instructions for:

while the second application view that corresponds to the second application is displayed:

displaying a fourth application view that corresponds to a fourth application, wherein the fourth application is in an active session, and wherein displaying the fourth application view includes displaying the first application view subsequent to the fourth application view in the order of application views, including:

in accordance with a determination that the fourth application was visited more recently than the second application, displaying the second application view subsequent to the fourth application view in the order of application views; and

in accordance with a determination that the second application was visited more recently than the fourth application, displaying the fourth application view subsequent to the second application view in the order of application views.

18. The non-transitory computer-readable storage medium of claim 13, wherein displaying the set of application views includes displaying the first application view without displaying a fifth application view that corresponds to a fifth application and wherein the one or more programs further include instructions for:

while displaying the first application view without displaying the fifth application view, detecting, via the one or more input devices, a first input; and
 in response to detecting the first input, displaying the fifth application view.

19. The non-transitory computer-readable storage medium of claim 13, wherein displaying the set of application views includes concurrently displaying the first application view and a sixth application view that at least partially overlaps the first application view.

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20. The non-transitory computer-readable storage medium of claim 13, wherein the one or more programs further include instructions for:

while displaying the first application view, detecting an input corresponding to a selection of the first application view; and

in response to detecting the input corresponding to selection of the first application view, displaying a user interface of the first application.

21. The non-transitory computer-readable storage medium of claim 13, wherein the first application view includes a representation of a user interface of the first application.

22. The non-transitory computer-readable storage medium of claim 13, wherein the request to display the set of application views includes a press of a physical button.

23. The non-transitory computer-readable storage medium of claim 13, wherein the request to display the set of application views is received while displaying a first respective user interface and wherein the one or more programs further include instructions for:

while displaying the set of application views, detecting, via the one or more input devices, a second input; and in response to detecting the second input, displaying the first respective user interface.

24. The non-transitory computer-readable storage medium of claim 13, wherein displaying the set of application views includes concurrently displaying in a second respective user interface:

a seventh application view in a first respective position in the order of application views; and an eighth application view in a second respective position in the order of application views, wherein the eighth application view is displayed at a location in a center of the second respective user interface.

25. A method, comprising:

at a computer system that is in communication with a display generation component and one or more input devices:

receiving, via the one or more input devices, a set of one or more inputs corresponding to a request to display a user interface of a respective application; in response to receiving the set of one or more inputs, displaying the user interface of the respective application;

while displaying the user interface of the respective application, receiving, via the one or more input devices, a request to display a set of application views; and

in response to receiving the request to display the set of application views, displaying, via the display generation component, the set of application views, wherein the set of application views includes a plurality of application views that includes a first application view that corresponds to a first application that is not in an active session and a second application view that corresponds to a second application, wherein:

the first application and the second application are different from the respective application;

an application that is actively performing one or more predefined functions is in an active session;

an application that is not actively performing one or more of the one or more predefined functions is not in an active session;

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the set of application views are arranged in an order of application views based on application activity status and application usage; and

displaying the set of application views includes:

in accordance with a determination that a second application is in an active session:

displaying the second application view that corresponds to the second application in a position before the first application view in the order of application views; and

displaying the first application view in a position that is subsequent to the second application view in the order of application views; and

in accordance with a determination that there are no applications in an active session, wherein the determination that there are no applications in an active session includes a determination that the first application is not in an active session: displaying the first application view in a position before the second application view in the order of application views; and displaying the second application view in a position that is subsequent to the first application view in the order of application views.

26. The method of claim 25, wherein displaying the second application view that corresponds to the second application includes displaying a status indicator that indicates a status of the second application.

27. The method of claim 25, further comprising:

while displaying the set of application views:

in accordance with a determination that a particular application is in an active session, displaying a status indicator that indicates a status of the particular application; and

in accordance with a determination that the particular application is not in an active session, displaying an application view that corresponds to the particular application without displaying the status indicator.

28. The method of claim 25, wherein displaying the set of application views includes displaying a third application view that corresponds to a third application, wherein displaying the third application view includes:

in accordance with a determination that the third application is in an active session, displaying the third application view in a first respective position in the order of application views; and

in accordance with a determination that the third application is not in an active session, displaying the third application view in a second respective position in the order of application views, wherein the second respective position in the order of application view is after the first respective position in the order of application views.

29. The method of claim 25, further comprising:

while the second application view that corresponds to the second application is displayed:

displaying a fourth application view that corresponds to a fourth application, wherein the fourth application is in an active session, and wherein displaying the fourth application view includes displaying the first application view subsequent to the fourth application view in the order of application views, including:

in accordance with a determination that the fourth application was visited more recently than the second application, displaying the second application view subsequent to the fourth application view in the order of application views; and

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in accordance with a determination that the second application was visited more recently than the fourth application, displaying the fourth application view subsequent to the second application view in the order of application views.

30. The method of claim 25, wherein displaying the set of application views includes displaying the first application view without displaying a fifth application view that corresponds to a fifth application and further comprising:

while displaying the first application view without displaying the fifth application view, detecting, via the one or more input devices, a first input; and
in response to detecting the first input, displaying the fifth application view.

31. The method of claim 25, wherein displaying the set of application views includes concurrently displaying the first application view and a sixth application view that at least partially overlaps the first application view.

32. The method of claim 25, further comprising:

while displaying the first application view, detecting an input corresponding to a selection of the first application view; and

in response to detecting the input corresponding to selection of the first application view, displaying a user interface of the first application.

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33. The method of claim 25, wherein the first application view includes a representation of a user interface of the first application.

34. The method of claim 25, wherein the request to display the set of application views includes a press of a physical button.

35. The method of claim 25, wherein the request to display the set of application views is received while displaying a first respective user interface and further comprising:

while displaying the set of application views, detecting, via the one or more input devices, a second input; and
in response to detecting the second input, displaying the first respective user interface.

36. The method of claim 25, wherein displaying the set of application views includes concurrently displaying in a second respective user interface:

a seventh application view in a first respective position in the order of application views; and

an eighth application view in a second respective position in the order of application views, wherein the eighth application view is displayed at a location in a center of the second respective user interface.

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